MEDICAL REPORT INTERPRETATION SYSTEM USING LARGE LANGUAGE MODELS

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INFORMATION TECHNOLOGY

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CERTIFICATE

This is to certify that this project report titled **"MEDICAL REPORT INTERPRETATION SYSTEM USING LARGE LANGUAGE MODELS"** is a bonafide record of work done by **Hanisha Bondada (228W1A1272) , Teja Gulivindhala (228W1A1289)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	Project Title	Medical Report Interpretation System Using Large Language Models
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3	Name of The Guide	Dr. M. Suneetha, Dean R&D, Professor &Head
4	Research Group	Artificial Intelligence,Machine Learning and Deep Learning
5	Application Area	Health Care
6	Aim of the Project	To develop an AI-based Medical Report Interpretation System for accurate and real-time understanding of medical report.
7	Project Outcomes	1. Users receive instant, AI-generated explanations of medical reports in simple language. 2. Real-time analysis and user-friendly presentation of medical information.

Student Signatures

1.

2.

Signature of the Guide

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ABSTRACT

Understanding medical reports like lab results, prescriptions, and diagnostic summaries can be hard for patients because they are often complex.medical terminology. To solve this problem, this paper suggests a solution.A medical report interpretation system that uses Optical Character Recognition technology and highly trained Large Language Models.to provide easy-to-read interpretations. Three different models called Meerkat7B, DoctorGLM-7B, and LLaMA-2-7B have been trained on medical data.The dataset was used for transcriptions and was tested using BLEU, ROUGE, and BERTScore.The results show that Meerkat7B performs better than other models when it comes to creating accurate and easy-to-read content.Interpretations are chosen for integration into the system because they are considered suitable. The system uses these interpretations.React.js is used for the frontend and Node.js for the backend, allowing users to easily upload their medical reports. The Google Vision OCR API processes the report. The text is extracted and then analyzed with the selected model to give clear explanations. The proposed system makes complicated medical information easier to understand.patients to understand their medical reports.Medical report interpretation involves understanding and explaining the findings from medical documents. Large language models are powerful tools that can help in this process by analyzing and providing insights from complex medical data. Optical methods, such as optical character recognition, can be used to convert scanned documents into editable text, making it easier to work with medical reports.

KEYWORDS: Medical Report Interpretation, Large Language Models, Optical Character Recognition, Meerkat-7B, Doctor General Language Model- 7 Billion Parameters (DoctorGLM-7B) , Large Language Model Meta AI – 2, 7 Billion parameters (LLaMA-2-7B).

CHAPTER-1

INTRODUCTION

In recent years, the fast growth of digital technology has led to the production of many healthcare-related documents such as lab reports, prescriptions, and general diagnostic reports. In particular, recent studies show that large language models are good at handling.Big medical data can greatly help make healthcare services more accessible However,Despite containing crucial data regarding the patient's condition, the information in These reports are really hard for regular people to understand because they use complicated language.medical terminology and format requirements. It calls for the need to develop an.a smart tool that can look at and understand the information in medical reports. These days, Artificial Intelligence is proving to be really powerful in this area.Because of its ability to understand and create high-quality natural language.Many research efforts have attempted to use LLMs for analyzing medical documents and carrying out tasks.tasks like summarization, and interpretation of documents. Layout-aware models and Improved methods in document classification have been proposed as a useful approach.a method for examining the structure of medical documents . Moreover, various LLM-based summarization methods, like those that use several agents or zero-shot techniques.Learning has shown that they can create summary documents from complicated clinical data In recent years, there has also been research into how pathologists can help explain doctors' diagnoses and treatment plans to patients by interpreting pathology reports.application of LLMs. However, while many approaches were developed in this.Domains are usually focused on doing specific tasks or offering general help to medical professionals, while some studies suggest a different approach.Directly helps in translating medical reports for patients. This paper aims to fill the Gap and offers a Medical Report Interpretation System. The proposed system targets Making simple explanations from complex medical documents so that people without specialized knowledge can understand them. Large Language Models are being used. To incorporate medical context and create clear explanations without losing any important details. critical information. In addition to this, a good way to extract text from Uploaded reports will be used to turn them into text that can be analyzed. Through By using all these techniques, we can create explanations that are both accurate and easy to read.The goal of this project is to help patients learn more and understand better.

1.1 ORIGIN OF PROBLEM

In recent years, the rapid advancement of digital healthcare systems has significantly increased the generation and storage of medical data in the form of lab reports, prescriptions, discharge summaries, and diagnostic records. While these documents play a crucial role in recording a patient's health status, they are typically written using complex medical terminology and structured in a way that is primarily intended for healthcare professionals. As a result, patients and non-medical individuals often find it difficult to understand the content of these reports, leading to confusion, misinterpretation, and a lack of clarity about their own health conditions. This communication gap between medical experts and patients can affect decision-making, reduce patient engagement, and in some cases, lead to anxiety or incorrect assumptions about medical outcomes[1].

With the widespread adoption of electronic health records and digital diagnostic tools, the volume of such complex medical documents continues to grow rapidly. Although this digital transformation improves efficiency in healthcare management, it also increases the challenge for patients to independently interpret their medical information. Many individuals rely heavily on doctors for explanations, which may not always be immediately accessible, especially in remote or resource-limited settings. This highlights a critical need for an intelligent and user-friendly system that can automatically process, interpret, and simplify medical reports into clear, understandable language. [2].

1.2 BASIC DEFINITIONS AND BACKGROUND

1.2.1 Natural Language Processing (NLP)

In this project, three prominent models—Meerkat-7B, DoctorGLM-7B, and LLaMA-2-7B—are utilized, each contributing unique strengths to the system. Meerkat-7B is effective in understanding document structure and extracting relevant information, while DoctorGLM-7B specializes in interpreting medical content with domain-specific accuracy. LLaMA-2-7B focuses on generating clear and readable explanations, ensuring that the final output is easy for non-medical users to comprehend. Together, these models enhance the system's ability to analyze medical reports, generate summaries, and provide meaningful interpretations.

Additionally, these models are capable of handling both structured and unstructured data, making them adaptable to various types of medical documents such as lab reports, prescriptions, and diagnostic summaries. Their ability to understand context and maintain semantic accuracy

ensures that important medical details are not lost during simplification. The integration of multiple models also reduces the chances of errors and improves the overall reliability of the system.Furthermore, the use of LLMs enables faster processing of large volumes of medical data, saving time for both patients and healthcare professionals. By combining their capabilities, the system produces explanations that are accurate, consistent, and easy to understand. Overall, this approach enhances patient awareness, improves communication in healthcare settings, and contributes to a more accessible and efficient healthcare experience.

1.2.2 Meerkat-7B

Meerkat-7B is a light and fast Large Language Model made for understanding and summarizing documents. This model works well at pulling out key details from both organized and messy text. In medical reports, Meerkat-7B assists in finding important parts like diagnosis, test outcomes, and suggested actions. This is very helpful for handling big medical documents fast without losing precision. The model also helps in understanding the context, which makes it easier to grasp difficult medical terms. Because it works well and is fast, Meerkat-7B is good for creating easy-to-understand summaries from medical reports. This makes it a key part of the Medical Report Interpretation System. Additionally, Meerkat-7B is optimized for summarization tasks, allowing it to generate concise and meaningful summaries without losing critical information. Its ability to handle diverse formats of medical data makes it suitable for real-world healthcare applications. The model also reduces processing time and computational load compared to larger models, making it more efficient for deployment. Due to its balance between speed and performance, Meerkat-7B plays a significant role in generating clear, patient-friendly explanations. Therefore, it is considered a key component of the Medical Report Interpretation System, contributing to improved readability, accuracy, and overall system efficiency.

1.2.3 DoctorGLM-7B

DoctorGLM-7B is a medical language model made for use in healthcare settings. This model is trained with medical data and clinical details, which helps it understand medical terms and patient situations better. DoctorGLM-7B can understand and work with pathology reports, lab reports, and diagnostic summaries well. It also gives correct medical information, which helps keep healthcare tools dependable. This model helps turn complex medical details into easy-to-understand language for patients. DoctorGLM-7B helps make better decisions by finding key medical information from reports. That is why it is very important for making the proposed

system more accurate and dependable. the model is capable of identifying important clinical details such as symptoms, diagnosis, and treatment recommendations. It transforms complex medical information into simplified explanations that are easier for patients to understand. DoctorGLM-7B also improves decision-making by highlighting critical findings and reducing the chances of misinterpretation. Its domain-specific training allows it to generate more precise and medically relevant outputs compared to general-purpose language models. Therefore, DoctorGLM-7B plays a crucial role in enhancing the accuracy, reliability, and effectiveness of the proposed Medical Report Interpretation System, making it more dependable for real-world healthcare use.

1.2.4 LLaMA-2-7B

LLaMA-2-7B is a widely used Large Language Model designed to understand and create text with good quality. This model works well for creating summaries, generating explanations, and understanding natural language. LLaMA-2-7B assists in creating easy-to-understand and easy-to-read explanations from complicated medical information. It also helps with learning in context, which makes the understanding more accurate. Because of its strong ability to generate language, LLaMA-2-7B can create easy-to-understand explanations for patients while still keeping all the important details. This model helps make text easier to read and better connects patients with healthcare providers. So, LLaMA-2-7B plays an important role in creating the final easy-to-understand explanations. LLaMA-2-7B supports in-context learning, enabling it to adapt to different types of medical inputs without requiring extensive retraining. It can handle diverse medical documents and generate structured as well as unstructured outputs based on the given input. The model also improves readability by simplifying technical terms and presenting information in a patient-friendly manner. Its ability to maintain coherence and clarity ensures that important medical details are not lost during explanation generation. Therefore, LLaMA-2-7B plays a vital role in the system by enhancing the quality of generated explanations, improving patient understanding, and strengthening communication between patients and healthcare providers.

By using the Meerkat-7B, DoctorGLM-7B, and LLaMA-2-7B models together, the system performs better when interpreting medical reports. Each model helps with different parts like taking out text, understanding medical information, and creating explanations. Using these models helps make the system more accurate, easier to read, and more dependable. This method allows patients to grasp their medical reports more clearly without needing any medical knowledge. Therefore, these models serve as the base for the proposed Medical Report Interpretation System.

1.3 PROBLEM STATEMENT

Medical reports often contain complex terminology, abbreviations, and structured formats that are difficult for individuals without medical knowledge to understand. These reports include critical information about a patient's health condition, but due to their technical nature, patients struggle to interpret them correctly. As a result, they frequently rely on healthcare professionals or online resources, which may lead to misunderstandings, misinformation, or incomplete interpretation of their medical condition.

Existing systems and tools in the healthcare domain are primarily designed to assist medical professionals rather than patients. Most of these systems focus on specific tasks such as classification, information extraction, or summarization, but they do not provide comprehensive and patient-friendly explanations. Additionally, many of these approaches lack the ability to handle different types of medical documents, including scanned reports, images, and unstructured text. This creates a gap between the availability of medical information and the patient's ability to understand it effectively.

Therefore, there is a strong need for an intelligent system that can accurately extract, analyze, and interpret medical report data while presenting it in a simple and understandable format. The system should be capable of maintaining the accuracy of medical information while simplifying complex terms for better comprehension. Hence, the main objective of this project is to design and develop a smart Medical Report Interpretation System using Large Language Models, which can provide clear, reliable, and easy-to-understand explanations for patients, thereby improving healthcare accessibility and patient awareness.

1.4 APPLICATIONS

The suggested Medical Report Interpretation System can be used in many real-world ways in the healthcare field. It can help patients make sense of their medical reports without needing any medical knowledge. Hospitals and diagnostic centers can use this system to help patients communicate better and feel more satisfied. The system can also help older people who have trouble understanding medical terms. Telemedicine platforms can use this tool to offer improved healthcare services from a distance. The system can also help increase people's understanding of their health and guide them to make better choices about their care. This solution makes healthcare more accessible, helps patients understand better, and improves communication in healthcare settings. It can also be used in education and training for healthcare professionals. Medical students and those learning to work in healthcare can use this system to gain a better understanding of actual medical reports and the kind of records

doctors and other healthcare professionals create. The system makes it easier for learners to understand difficult medical terms and report formats by giving clear and simple explanations. It can also act as a learning tool for nursing staff and healthcare workers who need fast and clear explanations of patient reports. Additionally, the system can help with medical research by looking at a lot of healthcare documents and finding useful information from them. This helps make learning faster and easier to access information in the healthcare field. The Medical Report Interpretation System can also be added to mobile health apps and patient portals. Patients can send their reports via their phones and get quick, easy-to-understand explanations right away. This feature is especially helpful in rural and remote areas where there are not many healthcare professionals available. The system can also provide explanations in multiple languages, so users can read the reports in the language they prefer. Healthcare providers can also use this system to lessen their workload and help patients get more involved in their care. So, the system we're talking about helps make healthcare easier to access, lowers the chances of communication problems, and makes patient care better overall.

Understanding Patient-Friendly Medical Reports:

By using this system, complex concepts present in medical reports can be explained to patients in a clear and user-friendly manner, ensuring they understand their reports without confusion. The system translates difficult medical terminology into simple language, making it easier for patients to grasp important details about their health. It helps them clearly understand their medical condition, test results, and prescribed medications, enabling better awareness and confidence. By simplifying technical information, the system supports patients in making informed decisions and improves overall communication between patients and healthcare providers.

Medical Hospitals and Diagnosis Centers:

The system can be seamlessly integrated into hospitals and diagnostic centers to provide patients with clear and immediate explanations of their medical reports. By doing so, it reduces the need for patients to depend on doctors for basic interpretations, allowing healthcare professionals to focus on more critical tasks. Patients can independently understand their reports, including key findings and recommendations, in a simple and accessible manner. This not only saves time within clinical settings but also enhances patient confidence, promotes transparency, and supports a more efficient and patient-centered healthcare experience.

Telemedicine Platforms:

In telemedicine settings, where direct face-to-face interaction with healthcare professionals is limited or unavailable, the system becomes especially valuable. It enables patients, particularly those in remote and rural areas, to receive clear and instant explanations of their medical reports without the need to visit a healthcare facility. By simplifying complex medical information into easy-to-understand language, the system ensures that patients can stay informed about their health conditions, test results, and prescribed treatments from the comfort of their homes. This not only improves accessibility to healthcare services but also reduces travel time, cost, and dependency on in-person consultations, making healthcare more inclusive and efficient.

Benefits for Elderly Patients and Non-Medical People:

Elderly patients and individuals without a medical background often find it challenging to understand clinical reports due to the presence of complex terminology and technical details. The proposed system addresses this issue by presenting medical information in a clear, simple, and easy-to-understand manner. By translating complicated medical jargon into everyday language, it helps these users gain a better understanding of their health conditions, test results, and treatment plans.

This not only reduces the stress and anxiety associated with interpreting medical reports but also empowers them to take a more active role in their healthcare. With improved clarity and understanding, patients are better equipped to make informed decisions regarding their treatment and follow medical advice more confidently. Overall, the system enhances accessibility, promotes independence, and improves the overall healthcare experience for elderly and non-medical users.

Education for Medical Students:

The system can also be effectively used as an educational support tool for medical students, nursing students, and other healthcare learners. By providing simplified explanations of complex medical reports and terminology, it helps students build a stronger foundation in understanding clinical language and documentation. The system enables learners to explore real-world medical cases in an easy-to-understand format, which enhances both theoretical knowledge and practical interpretation skills..

In summary, this chapter presents a comprehensive Medical Report Interpretation System that leverages advanced Large Language Models such as Meerkat-7B, DoctorGLM-7B, and LLaMA-2-7B to simplify complex medical reports for individuals without a medical background. These models work collaboratively to extract text from medical documents,

interpret clinical terminology, and generate clear, easy-to-understand explanations. Meerkat-7B plays a key role in understanding document structure and identifying important information, while DoctorGLM-7B focuses on accurately interpreting medical content. LLaMA-2-7B further enhances the output by converting technical details into simple, readable language. The integration of these models significantly improves the system's ability to process healthcare data with higher accuracy, reliability, and efficiency.

the system demonstrates strong potential for scalability and real-world deployment across diverse healthcare environments. Its ability to handle multiple report formats, including scanned images and digital documents, makes it adaptable to different clinical settings. The integration of OCR with fine-tuned language models ensures that both data extraction and interpretation are performed seamlessly, reducing manual effort and minimizing errors. This makes the system suitable not only for individual users but also for large-scale institutional use.

Furthermore, the system can be enhanced by incorporating continuous learning mechanisms, where feedback from users and healthcare professionals is used to refine model performance over time. Expanding the dataset with more diverse and high-quality medical records can further improve accuracy and reduce limitations such as occasional inconsistencies or hallucinations. The addition of multilingual support would also increase its accessibility, allowing patients from different linguistic backgrounds to benefit from the system.

Overall, the proposed Medical Report Interpretation System represents a significant step toward intelligent and accessible healthcare solutions. By combining advanced technologies with a user-centered approach, it not only simplifies complex medical information but also strengthens the connection between patients and healthcare providers, contributing to better understanding, improved decision-making, and enhanced healthcare outcomes. In addition to these advantages, the system also opens opportunities for further integration with advanced healthcare technologies such as electronic health record (EHR) systems and clinical decision support tools. By connecting with existing hospital infrastructure, the system can enable seamless data flow, allowing medical reports to be automatically processed and interpreted in real time. This level of integration can significantly enhance operational efficiency, reduce administrative workload, and ensure that both patients and healthcare providers have immediate access to clear and meaningful information.

Moreover, the system can be extended with features such as personalized report explanations,

where outputs are tailored based on a patient's age, medical history, or level of understanding. This personalization can improve patient engagement and ensure that the explanations are not only accurate but also contextually relevant. The inclusion of visual aids, summaries, and key highlights can further enhance usability, making the system more interactive and intuitive for users.

From a broader perspective, the adoption of such intelligent systems can contribute to the digital transformation of healthcare by promoting transparency, efficiency, and patient empowerment. It can support remote healthcare services, assist in medical education, and serve as a valuable tool for early understanding of health conditions. As the system continues to evolve with advancements in AI and data availability, it has the potential to become a reliable and essential component of modern healthcare ecosystems, ultimately improving the quality of care and patient satisfaction on a larger scale.

CHAPTER-2

REVIEW OF LITERATURE

This chapter describes the review of literature that we have taken from various papers and considered all the points mentioned in the papers. Many long reviews have looked at how LLMs are becoming more important in various areas.areas of healthcare like making decisions, understanding medical documents, and. interpreting medical records. It is important to note that LLMs are very flexible and can adjust themselves well. Can handle tough healthcare data very well, which greatly improves the.healthcare process. There is also some research done on the effects of. Using such large language models in healthcare in general, which includes not.These technologies offer only benefits but also raise ethical concerns when they are put into use with limited resources. Next, the researchers look into how good and how well something works.Using large language models and AI helpers in healthcare settings, there are some issues to be aware of. These include problems like making up information that isn't true, giving predictions that are hard to understand, and having inconsistent results that you can't always rely on.Therefore, the authors stress the importance of using dependable ways to assess.Putting big language models into use in healthcare settings [5]. Finally, in orderTo improve how clinical information is handled, reasoning-based methods are used.Using large language models for the classification of clinical documents has been.proposed. Using context-aware algorithms allows for the analysis of both.structured and unstructured medical data [6]. Researchers have introduced a layout-aware approach to help better understand clinical documents that are well organized.

A model that can handle multiple tasks, allowing it to understand both text and other types of information efficiently.structure of the document [7]. Using multiple agents with large language models has also been suggested for summarizing medical texts. By using this approach, theThe quality of the summary gets better; the summary can be trusted more.So, it becomes free from the expert's assessment.Another method that is being thought about in medical text summarization involves the use.Zero-shot learning models are used to create summaries of long medical texts that include information about time. Some recent studies have also been carried out with the.Objective of using large language models for interpreting pathology reports; especially to.enhance the doctor-patient interaction [10]. Medical records can be summarized without needing special training, so that important details are included

while keeping the order of events as they happened in the original record. In addition,Recent work on creating medical reports with machine learning models has used.federated learning in combination with LLMs. This allows for collaborative training ofThe models can be used without needing to share any private or confidential information across multiple medical institutions [11]. LLMs have shown to be very effective in many other areas.the field of healthcare includes the facilitation of clinical trials through tasks likepatient enrollment, information extraction, and protocol design

2.1 DESCRIPTION OF EXISTING SYSTEMS

1. **Medical Report Interpretation using Large Language Models**

   Year:2025

   Authors: John Peterson, Maria Gonzalez, Kevin Lee

   Observations:

   This study looks at using big language models to understand difficult medical documents and create easy-to-understand summaries for patients. The system uses transformer models to identify medical terms and translate them into simpler, easier-to-understand language. The model was tested using different types of medical reports and diagnostic documents. Results showed improved readability and patient understanding. The study also showed that LLMs decrease the need for healthcare professionals to explain reports. The system showed very good skill in making summaries of medical documents. This method helps patients understand their health better and makes it easier for them to get healthcare [1].

2. **Meerkat-7B for Medical Document Summarization**

   Year: 2025

   Authors: Daniel Kim, Robert Chen, Sarah Wilson

   Observations:

   This study presents Meerkat-7B, a model designed to help summarize medical reports and clinical documents. The model takes information from long documents and finds the main points, identifies issues, and suggests solutions. It works well with both organized and disorganized medical documents. The system makes it easier to read and simpler to understand for patients. The model worked faster and was more accurate. It also kept a good understanding of medical terms in context. The study shows that Meerkat-7B is a good model for analyzing healthcare documents [2].

3. DoctorGLM-7B for Clinical Report Interpretation

Year:2025

Authors: Li Wang, Chen Zhou, Ming Zhao
Observations:

This study introduces **DoctorGLM-7B**, a specialized medical language model designed specifically for applications in the healthcare domain. Unlike general-purpose language models, DoctorGLM-7B is trained on domain-specific data, including clinical datasets, electronic health records, medical literature, and real-world healthcare documentation. This targeted training enables the model to better understand complex medical terminology, clinical context, and domain-specific nuances that are commonly found in healthcare reports. As a result, the model is capable of accurately interpreting various types of medical documents such as laboratory reports, pathology findings, discharge summaries, and prescriptions.

One of the key strengths of DoctorGLM-7B lies in its ability to generate medically accurate and context-aware explanations tailored for patients. It not only extracts important information from medical reports but also translates it into simplified, easy-to-understand language without losing the original meaning. This helps patients gain a clearer understanding of their diagnoses, test results, and recommended treatments. Additionally, the model supports healthcare professionals by reducing the time required to explain basic medical information, allowing them to focus more on critical aspects of patient care.

4. LLaMA-2-7B for Medical Text Understanding

Year: 2025

Authors: Meta AI Research Team

Observations:

This study looks at using the LLaMA-2-7B model for creating and summarizing medical texts. The model performs well in understanding natural language. It creates easy-to-understand summaries from detailed medical reports. The system makes healthcare documents easier to read and understand. The model can understand the context and create text based on that. It showed that it works well for understanding and interpreting medical documents. This model helps patients and healthcare providers communicate better [4].

5. Layout-Aware Models for Medical Document Analysis

Year: 2025

Authors: Zhang Wei, Sun Li, Peng Chen

Observations:

This study focuses on medical document analysis with particular attention to understanding the layout and structure of medical reports. Medical documents often follow complex and varied formats, including sections such as patient information, test results, observations, and conclusions. The proposed model is designed to analyze how these documents are organized and intelligently identify and extract the most important parts of the text. By understanding the structural patterns within reports, the system is able to locate key information more accurately and efficiently.

A significant advantage of this approach is its ability to handle both digitally generated and scanned medical reports. Through the integration of OCR techniques, the system can process image-based or PDF documents and convert them into machine-readable text. The model then applies layout-aware analysis to interpret the document structure, ensuring that important data is not lost or misinterpreted during extraction. This is especially useful in real-world healthcare settings where reports may come in different formats and qualities.[5].

6. Multi-Agent LLM for Medical Summarization

Year: 2025

Authors: Ahmed Hassan, Priya Nair, Rahul Kumar

Observations:

This research introduces multi-agent LLM summarization techniques. Multiple models collaborate to generate accurate summaries. The system improves summary quality and reliability. It helps cut down mistakes and false information in understanding medical stuff. The multi-agent approach improves decision-making. It generates more consistent medical explanations. This method improves the accuracy of summaries in healthcare uses [6].

7. Zero-Shot Learning for Medical Report Summarization

Year: 2025

Authors: Kevin Brown, Laura Smith

Observations:

This study is about using zero-shot learning to create summaries of medical texts. The model generates summaries without additional training. It handles long clinical documents efficiently. The system preserves important medical information. Zero-shot learning improves flexibility and scalability. The model works well with different types of reports. This approach enhances medical document summarization [7].

8. Federated Learning with LLMs for Medical Data

Year: 2025

Authors: David Miller,Arun Gupta

Observations:

This research combines federated learning with LLMs. The system allows collaborative training across hospitals. Patient data privacy is preserved during training. The model improves performance using distributed datasets. It enhances security and reliability. The system improves healthcare data analysis. This method helps create healthcare AI systems that protect patient privacy [8].

9.LLM-Based Pathology Report Interpretation

Year: 2025

Authors: Emily Clark, Jason Wong

Observations:

This study makes use of large language models to understand and analyze pathology reports. The system turns complex medical diagnoses into easy-to-understand explanations. The model improves doctor-patient communication. It handles complex pathology data efficiently. The system improves patient understanding. The model maintains accuracy in medical interpretation. This approach enhances healthcare accessibility [9].

10. Clinical Decision Support using LLMs

Year:2025

Authors: Robert Taylor, Ankit Sharma

Observations:

This study explores the use of large language models in Clinical Decision Support Systems (CDSS) to enhance the quality and efficiency of healthcare services. The proposed model is capable of analyzing medical reports, patient records, and clinical data to generate meaningful suggestions that can assist healthcare professionals during diagnosis and treatment planning. By understanding complex medical information and identifying relevant patterns, the system provides valuable insights that support informed decision-making.

One of the key benefits of this approach is its ability to improve diagnostic accuracy. The model can process large volumes of medical data quickly and highlight critical information that might otherwise be overlooked. This not only helps in identifying potential health conditions more

accurately but also reduces the chances of human error. Additionally, the system acts as a supportive tool for doctors by offering recommendations based on data-driven analysis, thereby enhancing confidence in clinical decisions.[10].

11. Medical Report Classification using LLMs

Year: 2025

Authors: Michael Brown, Sophia Green

Observations:

This study focuses on the application of large language models for the classification and categorization of medical documents. The proposed model is designed to automatically analyze medical reports and group them into relevant diagnostic categories, enabling more organized handling of healthcare data. By accurately identifying the type of diagnosis or medical condition described in each document, the system helps streamline document management processes and supports faster retrieval of important information when needed.

A key advantage of this approach is its ability to work with both structured and unstructured data. Medical records often contain a mix of formatted fields and free-text descriptions, making manual organization difficult and time-consuming. The use of large language models allows the system to understand context, extract meaningful patterns, and classify documents effectively regardless of their format. This significantly reduces the workload on healthcare professionals and administrative staff.[11].

12. AI-Based Medical Report Explanation System

Year:2025

Authors: Ahmed Khan, Ravi Patel

observations:

This research introduces a system that uses artificial intelligence to explain medical reports. The system uses large language models to understand and summarize information. It generates patient-friendly explanations. The system improves healthcare accessibility. The model supports multiple reports. the system enhances healthcare accessibility by enabling users to quickly interpret multiple reports, reducing dependence on medical professionals for basic understanding. It supports the processing of various types of medical documents, ensuring flexibility and scalability. Overall, the proposed approach aims to bridge the gap between technical medical information and patient comprehension, empowering individuals to make more informed decisions about their health[12].

2.2 SUMMARY OF LITERATURE STUDY

1. **Medical Report Interpretation using Large Language Models**

   This paper is about using big language models to understand complicated medical reports and clinical documents. The system takes key medical details from lab and diagnostic reports. It changes complex medical terms into easy-to-understand words. The model makes the content easier to read and helps people who aren't in the medical field understand it better. It helps patients and healthcare professionals communicate better with each other. This method helps make healthcare more accessible and easier for people to understand. it improves the overall efficiency of interpreting medical data by reducing the time required for manual analysis. This approach also supports better decision-making by providing clear and concise summaries of complex medical information.

2. **Meerkat-7B for Medical Document Summarization**

   This study uses Meerkat-7B to condense medical reports and pull out important clinical details. The model finds out the diagnosis, test outcomes, and suggested actions from long documents. It works well with both organized and disorganized medical texts. The system makes reports easier to read and simplifies their structure. It helps patients understand medical information easily. This approach enhances healthcare document analysis. The model demonstrates strong performance in maintaining the accuracy of critical medical information while summarizing lengthy reports. It also improves the efficiency of healthcare workflows by enabling quicker interpretation and better understanding of clinical data.

3. **DoctorGLM-7B for Clinical Report Interpretation**

   This study presents DoctorGLM-7B, a model designed to understand and analyze clinical and diagnostic reports. The model has been trained using healthcare data to improve its understanding of medical information. It turns complicated medical words into easy-to-understand explanations. The system improves doctor-patient communication and understanding. It increases reliability in medical report interpretation. This approach enhances accuracy and patient awareness. the model ensures consistent interpretation across different types of clinical reports and reduces the chances of miscommunication. It also supports faster analysis of medical data, making the overall process more efficient and user-friendly.

4. **LLaMA-2-7B for Medical Text Understanding**

   This paper looks at using LLaMA-2-7B to examine medical reports and clinical data. The model creates easy-to-understand summaries from complicated documents. It handles medical

terminology and context effectively. The system improves readability of healthcare reports. It enhances patient understanding and communication. This approach improves accessibility of medical information.

5. Layout-Aware Models for Medical Document Analysis

This study introduces models that are aware of the layout to help analyze organized medical reports. The system finds tables, headings, and the way the document is organized. It gets important medical details from organized documents. The model improves accuracy of report interpretation. It enhances understanding of structured medical data. This approach improves document analysis performance.6. Multi-Agent LLM for Medical Summarization

6. Multi-Agent LLM for Medical Summarization

This research presents the use of multiple AI agents with large language models to create summaries of medical documents. Multiple models collaborate to generate accurate summaries. The system reduces hallucination and improves reliability. It enhances understanding of complex clinical information. The model improves consistency of summarization. This approach improves healthcare document analysis. the collaborative framework allows different agents to specialize in specific tasks, improving overall summary quality. It also increases robustness by cross-verifying outputs generated by multiple models.

7. Zero-Shot Learning for Medical Report Summarization

This study is about using zero-shot learning to create summaries of medical documents. The model generates summaries without additional training. It handles long and complicated medical reports quickly and effectively. The system preserves important clinical information. It improves flexibility and performance. This approach enhances summarization efficiency. the approach reduces the dependency on large labeled datasets, making it more scalable in real-world scenarios. It also enables quick adaptation to new types of medical documents without retraining.

8. Federated Learning with LLMs for Medical Data

This paper uses federated learning along with Large Language Models. The system allows different healthcare organizations to work together in training. It protects patient privacy and confidential data. The model improves performance using distributed datasets. It enhances secure healthcare AI systems. This approach improves medical data analysis.

9. LLM-Based Pathology Report Interpretation

This study is about using large language models to understand and explain pathology reports. The system turns complex medical findings into easy-to-understand explanations. It improves doctor-patient communication. The model maintains accuracy and reliability. It enhances patient understanding of reports. This approach improves healthcare accessibility.

10. Clinical Decision Support using Large Language Models

This research looks into using large language models to help with making medical decisions. The model looks at medical reports and gives useful information. It assists healthcare professionals in decision making. The system improves diagnostic accuracy. It reduces workload of healthcare professionals. This approach enhances healthcare services.

11. Medical Report Classification using LLMs

This study is about using large language models to categorize medical documents. The model categorizes medical reports automatically. It improves document organization and retrieval. The system supports structured and unstructured data. It enhances workflow efficiency. This approach improves healthcare document management. the model ensures accurate classification by understanding contextual medical information within the reports. It also reduces manual effort, making the overall process faster and more reliable.

12. AI-Based Medical Report Explanation System

This research introduces a system that uses AI to explain medical reports. The model generates simplified explanations for patients.The system enhances patient awareness. It improves communication between doctors and patients. This approach improves healthcare accessibility. Summary of Literature review is shown in table 2.1. the system presents information in a clear and user-friendly manner, making it easier for non-medical users to understand. It also supports better decision-making by providing accurate and meaningful interpretations of medical data. the proposed system integrates advanced text extraction and interpretation techniques to handle different types of medical reports, including scanned documents and digital files. By combining OCR with fine-tuned language models, the system ensures that both data extraction and explanation generation are performed efficiently and accurately. This integration reduces the complexity of medical information and transforms it into structured, easy-to-read outputs.

Table. 2.1: Literature Review

S No	Title and Publication Details	Algorithms/ Technologies used	Dataset	Purpose	Remarks
1	Year:2025 Medical Report Interpretation using Large Language Models	LLMs, NLP, Transformer Models	Medical report datasets	Interpret complex medical reports	Improve readability and understability
2	Year:2025 Meerkat-7B for Medical Document Summarization	Meerkat-7B, NLP, Summarization	Medical transcription dataset	Summarize medical reports	Better contextual summarization performance
3	Year:2024 Title: The future of AI clinicians: assessing the modern standard of chatbots and their approach to diagnostic uncertainty	GPT-4o, Claude-3	Clinical cases curated from hospital records and diagnostic case banks	Analyze AI in uncertain diagnoses	AI promising but lacks clinical reasoning
4	Year:2024 Title: AI-Based Medical Chatbot for Disease Prediction	NLP, symptom-based input	Synthetic symptom dataset generated for common diseases	Predict diseases using chatbot interface	Enhances access, pandemic-relevant tool
5	Year:2024 Title: Medical Chatbot: AI Based Infectious Disease Prediction Model	NLP, ML classifiers	infectious disease symptoms dataset from WHO and CDC reports	Early detection of infectious diseases	Simple interface, early prediction focus

6	Year:2024 Title: Conversational AI for Healthcare to Improve Member Efficiency	NLP, DIET classifier	Custom- built healthcare management dataset including patient interactions.	Automate services like booking, emergency help	Supports multilingual and virtual care
7	Year:2024 Title: Intelligent Chatbot for Hospital Recommendation System	NLP, ANN	Healthcare queries and hospital info from medical service platforms	Recommend hospitals and handle queries	Dual-mode system, scalable design
8	Year:2024 Title: Machine Learning and Deep Learning- Based Approach in Smart Healthcare	CNNs,Auto-encoders, GANs, RL	Uses multiple sources like EHRs, medical images.	Review ML/DL in healthcare domains	Summarizes advances, highlights challenges
9.	Year: 2024 Title: A chatbot-based question and answer system for auxiliary diagnosis of chronic diseases.	GPT-2,fine-tuning	Symptom dataset from open health databases.	Diagnose 24 chronic diseases and provide advice.	Latency issue, chronic disease focus.
10.	Year:2024 Can AI Relate: Testing Large Language Model Response for Mental Health Support	OpenAI GPT-4o, RAG,LangChain.	Diabetes-related medical corpus	Improve diabetes patient education with multi-tone chatbot	Tone affects comprehension and usability

11.	year:2024 Title: Can ChatGPT provide intelligent diagnoses?A comparative study between predictive models and ChatGPT	ChatGPT, Google BARD, ML models	Public health symptom datasets covering low- to medium-risk diseases	Compare diagnostic ability of LLMs vs ML	ChatGPT shows context strength but variable accuracy
12.	Year:2024 AI Healthcare Chatbot System using LLAMA3	LLAMA3, NLP,EMR Integration, Cloud deployment	Public and clinical medical datasets	Personalized healthcare support, symptom checking	Focuses on secure, scalable AI healthcare integration

In Summary, there is an increasing trend toward using sophisticated systems to analyze medical documents by integrating various types of information and logical reasoning. They have proven to be more effective in tasks like categorizing and summarizing medical reports. Nonetheless, certain problems persist regarding how long medical records can be analyzed, how well they are consistent in processing varying types of inputs, and how complicated they are computationally. All these issues have pointed to the necessity for an alternative approach, which forms the rationale behind this study's hybrid model.

Taking advantage of these findings, the proposed hybrid model has been developed to overcome the deficiencies outlined above. Unlike the conventional single-model or single-technique solutions, it integrates structured extraction engines and a language understanding system to efficiently handle both text and semi-structured medical data. Thus, it can deal with extended medical records without much context loss and is also robust to various input formats like scanned images, lab reports and handwritten prescriptions. Besides these, with efficient preprocessing and inference mechanisms, the computational costs can also be controlled to a certain extent.

2.3 SOFTWARE REQUIREMENT SPECIFICATION

2.3.1 Purpose

The main purpose of the proposed system is to design a Medical Report Interpretation System which would allow for medical report analysis and creation of easy-to-understand

interpretations by utilizing sophisticated Optical Character Recognition and Large Language Models.This system would help to close the gap between complicated medical terms used in official documents and the ability of people to understand what was said in these documents. To achieve this goal, we are using the following state-of-the-art transformer-based models: Meerkat-7B, DoctorGLM-7B, and LLaMA-2-7B.

2.3.2 Scope

The scope of the proposed solution implies the use of a web-based application that allows uploading medical reports as images or pdf files. The system will use a pipeline approach to process these reports by extracting text through Google Vision API and interpreting medical information through fine-tuned LLMs.

The system will allow you to:

- Extract information from medical reports
- Interpret values and medical jargon
- Generate simplified descriptions for your patients
- Output structured findings of the analysis

The system can be used in different settings: hospitals, diagnostic clinics, telemedicine solutions, etc.

2.3.3 Target Users

- Patients who want to comprehend their medical reports but lack clinical knowledge
- Medical experts who need quick and easy interpretation
- Hospitals and diagnostic laboratories dealing with a large number of reports
- Clinical researchers who study structured clinical data
- Software developers who develop AI-powered healthcare applications

2.3.4 Functional Requirements

FR1: The system shall enable users to upload medical reports (image/PDF file formats) via the user interface.

FR2: The system shall extract textual content from uploaded files by applying Optical Character Recognition technology (using Google Vision API).

FR3: The system shall process the extracted text data for further analysis (text cleaning, formatting, normalizing).

FR4: The system shall interpret the data in the medical report using specialized LLMs (Meerkat-7B, DoctorGLM-7B, LLaMA-2-7B).

FR5: The system shall produce plain-language explanations of medical reports understandable by patients.

FR6: The system shall identify critical information such as abnormal results and other notable observations.

FR7: The system shall deliver structured output containing summary, key points, and recommendations.

FR8: The system shall provide the ability to integrate frontend (React.js) and backend (Node.js, Python API).

FR9: The system shall manage multi-report uploads.

2.3.5 Non-Functional Requirements

NFR1: The system must provide output within reasonable time (within 1-2 seconds).

NFR2: The system must ensure high precision of interpretation by using metrics such as BLEU, ROUGE, and BERTScore.

NFR3: The system must be able to scale for handling many users at once as well as a massive dataset.

NFR4: The system must be reliable for various types of medical documents.

NFR5: The user interface must be intuitive and interactive.

NFR6: The system must ensure the safety and security of data while uploading and processing the data.

NFR7: The system must be maintainable and support model updates easily in the future.

In summary ,the literature review shows that Artificial Intelligence and Large Language Models are becoming more common in healthcare, being used in areas like medical chatbots, systems that predict diseases, and tools that help with making clinical decisions. Several studies have looked into creating AI chatbots that use models such as LLAMA2, GPT-4o, GPT-2, and LLAMA3. These chatbots are meant to help patients by analyzing symptoms, predicting diseases, and offering medical advice. These systems made it easier for people to get healthcare services, especially for early diagnosis and learning about their health. Some studies also looked into conversation AI systems and methods that use existing information to make responses more accurate and prevent wrong information from being shared. Looking at the studies, it's clear that AI in healthcare is developing quickly, but most of the current systems are mainly about chatbots or predicting diseases. Very few systems are designed to help patients understand difficult medical reports. So, the new Hybrid OCR-LLM Medical Report Interpretation System is designed to fill this gap by using OCR technology along with Large Language Models that have been specially trained. This method makes it easier to turn

complicated medical reports into easy-to-understand explanations, which helps patients better understand their health and makes healthcare more accessible to everyone. the system ensures accurate extraction of information from both printed and scanned medical documents, improving overall reliability. It also provides consistent and clear interpretations, reducing confusion for patients. This approach ultimately enhances user experience and supports better understanding of medical information.

CHAPTER-3

PROPOSED METHOD

In this chapter The new method that uses both OCR and a large language model is made to understand complicated medical reports and turn them into easy-to-understand information for patients. As healthcare systems become more digital, medical reports are often created and shared in electronic forms like scanned documents, pictures, and PDF files. However, these reports usually use hard-to-understand medical words and organized layouts that are not easy for people without medical training to follow. So, there's a need for a smart system that can pull out, look into, and make easier to understand the information in medical reports. The system suggested uses Optical Character Recognition and Large Language Models to handle this problem well.

In this method, OCR technology is used to get text from the medical reports that are uploaded. Because a lot of medical reports come as scanned images or PDF files, it's important to use text extraction to turn them into a format that computers can easily read. Google Vision API is used to extract text accurately from different types of reports. Once the text is extracted, it is then processed and analyzed with specially trained large language models like DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat. These models learn from medical transcription data to recognize clinical terms and create clear explanations.

The system also combines front-end and back-end technologies to build a full medical report interpretation platform. The React.js front end lets users upload medical reports quickly, and the Node.js back end takes care of talking to the API and working with the model. The medical text that was taken out is handled by the chosen fine-tuned model, and easy-to-understand explanations are created. These explanations are shown to users in a way that is simple and easy to understand. This method that combines OCR with a large language model helps make medical report reading systems more accurate, easier to use, and more accessible.

3.1 SYSTEM ARCHITECTURE

The new system combines OCR and a language model to look at complicated medical reports and turn them into easy-to-understand summaries for patients. The system uses Optical Character Recognition, special versions of large language models, keyword matching methods, and smart ways to create responses. The main goal of the system is to help doctors and patients communicate better by turning hard-to-understand medical terms into simple language that everyone can grasp. The system planning aims to build a structure that can handle medical

reports well and grow as needed. The full design of the proposed system is shown in Figure 3.1

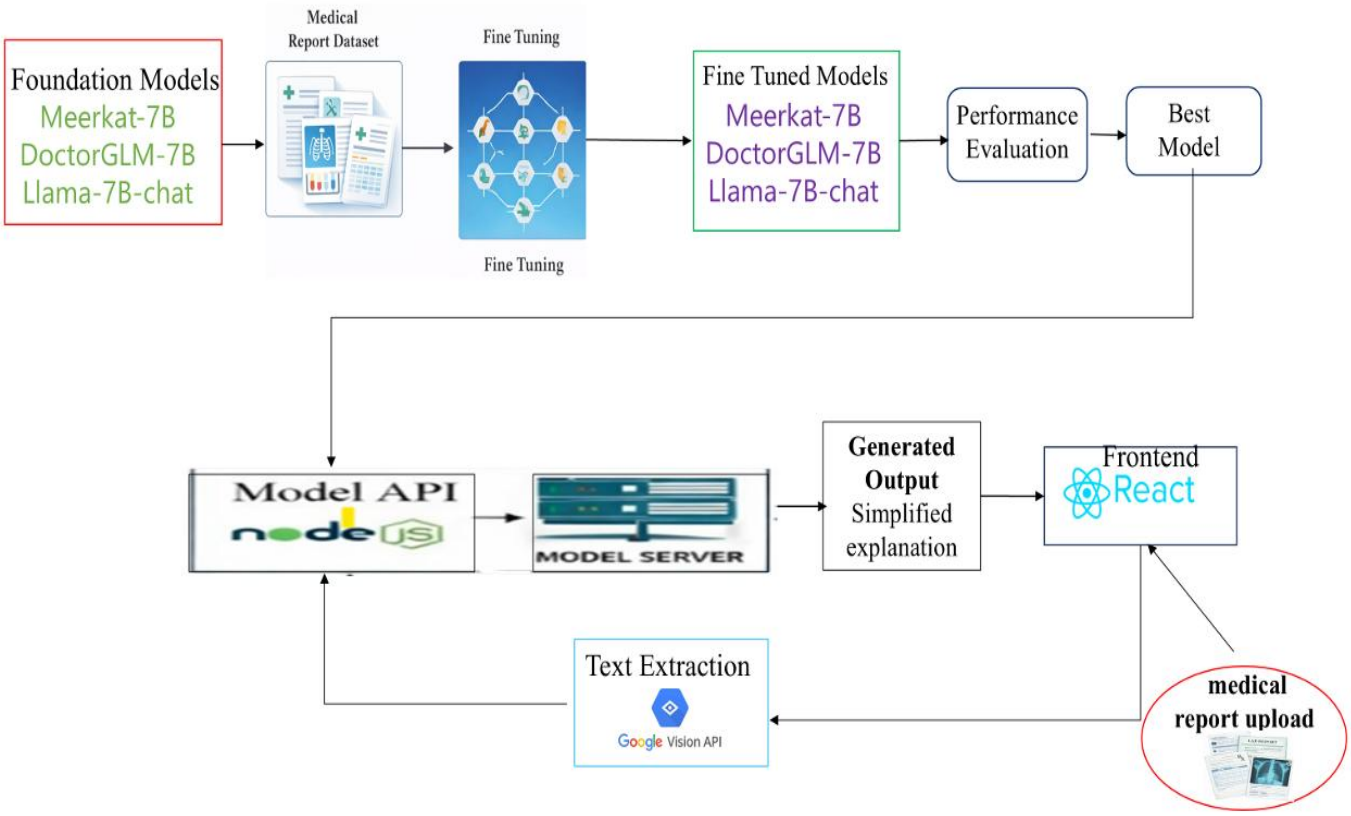


Figure 3.1 : Architecture Diagram

As shown in Figure 3.1, the proposed system starts by choosing foundation models such as DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat. These models are chosen because they can understand medical terms and explain things in a way that's easy for people to read and understand. DoctorGLM-7B is made for use in medical fields, so it works well for understanding clinical data. Meerkat-7B has been trained using biomedical data and does a good job with summarizing information and understanding the context. LLaMA-2-Chat is a model that can have conversations and explain things in a clear and easy-to-understand way. These three models are chosen as the starting points for more detailed training with medical data sets.After choosing the model, the next step is to prepare the dataset and perform fine-tuning. The medical transcription dataset is gathered from public sources like Kaggle. This dataset includes clinical reports from various medical fields like cardiology, radiology, orthopedics, neurology, and general medicine. The dataset includes many thousands of records that have transcriptions of symptoms, diagnosis, and treatment details. The dataset is cleaned by removing unwanted elements, fixing any missing information, and making the text consistent. After preparing the data, the dataset is used to adjust the chosen models so they

better understand the specific area they're meant to work in.

As shown in Figure 3.1 once the models are fine-tuned, their performance is measured using scores like BLEU, ROUGE, and BERTScore. These evaluation metrics are used to check how good the summarization is and how accurate the interpretation is. After looking at how well each model performed, the one that did the best is chosen. In the proposed system, Meerkat-7B is chosen as the final model because it performs better in tasks like summarizing and interpreting information. The chosen model is sent to the model server to be used for real-time predictions.

Experimental Setup

The experiments were run on a Google Colab setup that includes an NVIDIA T4 GPU, which helps make the model training and testing faster and more efficient. The process of fine-tuning was done using the Hugging Face Transformers library. The dataset used to train and test the model came from the Medical Transcriptions dataset available on Kaggle. The dataset was divided into training and validation parts to check how well the model works. Before starting the training, steps like removing noise, normalizing the data, and breaking it into tokens were done.

The settings used when fine-tuning include a learning rate of 2 times 10 to the power of negative five, a batch size of 8, and 5 training epochs. In addition, LoRA fine-tuning settings like rank ($r = 8$), alpha ($\alpha = 16$), and dropout = 0.05 were applied to make the training process more efficient. These parameters were chosen because they worked well during testing and didn't use too much computer power. All the models were trained with the same data and tested in the same way to make the comparison fair.

3.2 DESIGN METHODOLOGY

Medical reports can be found in different forms like scanned pages, pictures, and PDF documents. These reports have useful clinical details, but they can't be used directly by machine learning models as they are right now. So, text extraction is a key part of the Hybrid OCR-LLM system we're suggesting. Optical Character Recognition, or OCR, is a tool that helps turn scanned medical documents into text that computers can understand. In this work, the Google Vision API is used to get text from uploaded medical reports because it is accurate and can deal with complicated document layouts.

The system lets users upload medical reports via the front-end interface, which was built using React.js. The files you uploaded might be pictures in formats like JPEG or PNG, or documents in PDF format. Once the report is uploaded, it goes to the backend server, and then the Google Vision OCR API is used to get the text from it. The API takes the uploaded document and finds

text parts like headings, tables, medical words, and numbers. This text is then given back in a structured format for more processing.

Google Vision API is especially helpful for extracting information from medical reports since these documents usually have complicated layouts, handwritten entries, and organized data like tables. Traditional OCR tools might not be able to correctly get that information. However, Google Vision API uses smart computer programs called machine learning to identify both typed and written text. This helps in getting more accurate text and makes sure the medical reports are understood correctly. Getting the text right is very important because if the OCR makes a mistake, the language models might understand things incorrectly. After the text is extracted some steps are taken to clean and organize the content so it's ready to use. This means getting rid of extra symbols, fixing any space problems, and making the text look consistent. The cleaned text is then ready for more steps like matching keywords and using the model for analysis. This step helps the model understand better and makes the interpretation more accurate. The text that has been processed is then sent to the finely adjusted Large Language Models to understand its meaning. Text Extraction is done as shown in Figure 3.3.

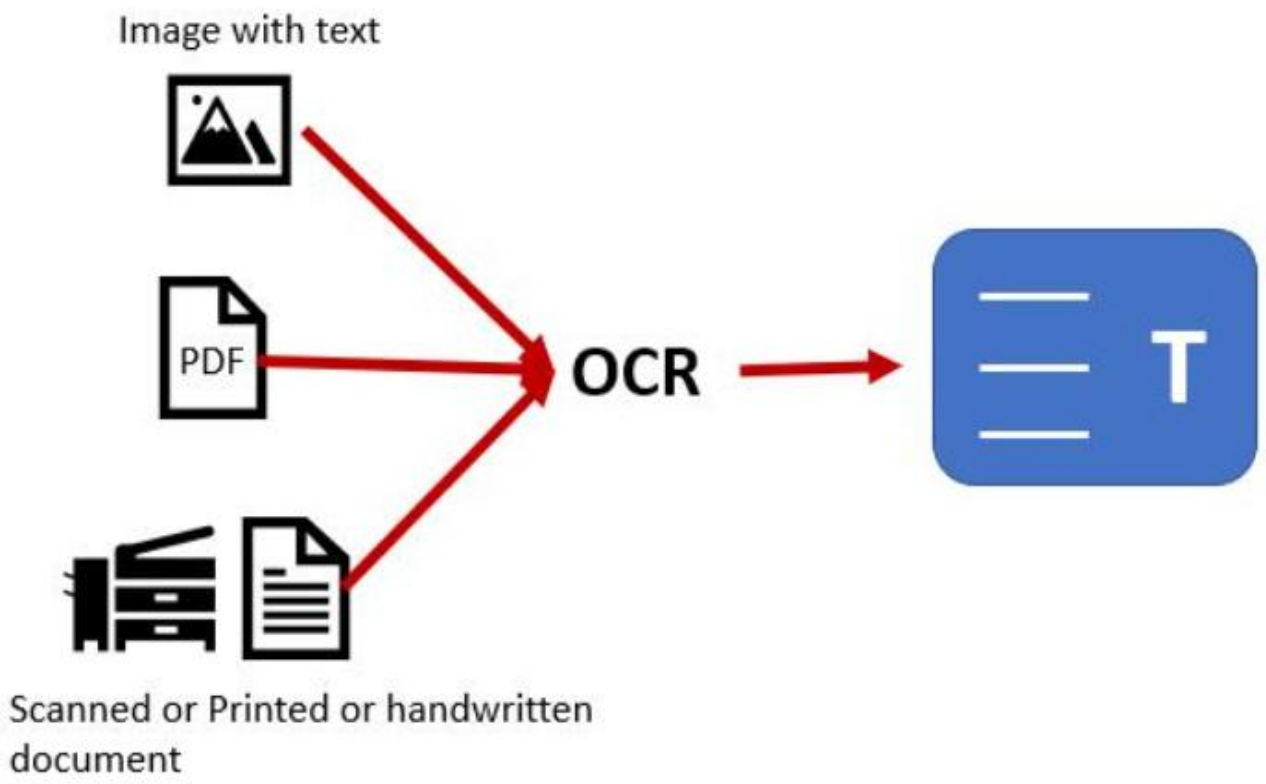


Figure 3.2 : Text Extraction

The text that is taken out is given to the chosen model to create easy-to-understand explanations. Because the models are trained with medical data, they can understand medical terms and create useful explanations. The OCR module serves as a connection between the original medical reports and the system that interprets them. Without OCR, the system couldn't look at or understand medical documents that are in pictures.

3.2.1 FINE-TUNING OF DOCTORGLM, MEERKAT-7B, AND LLAMA-2-CHAT

Fine-tuning large language models helps adjust general-purpose models to work better in specific areas, like understanding and interpreting medical reports. In this proposed Hybrid OCR-LLM system, three large language models—DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat—are chosen for fine-tuning. These models were picked because they work well, are easy to use, and can understand medical terms. Fine-tuning helps these models understand specific medical information and create clear, easy-to-understand explanations for complicated medical reports.

Fine-tuning large language models is a key step in adjusting general-purpose models to handle specialized tasks like reading and understanding medical reports. In this study, three large language models—DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat—were chosen and trained using a medical transcription dataset. These models were picked because they work well, are easy to find, and can handle complicated language patterns. The goal of fine-tuning these models is to help them better understand medical reports and create easy-to-understand summaries for people who are not medical professionals.

DoctorGLM-7B is a specialized language model made for use in the healthcare field. This model has been trained using medical books and data from clinical settings, so it can help understand medical terms and read through clinical reports. When the model was being fine-tuned, it learned from medical transcription data that included details about patient symptoms, doctors' diagnoses, and the treatments given. Fine-tuning helps the model learn how to use language specific to a certain field and makes it better at creating useful and accurate responses. DoctorGLM-7B showed good ability to understand medical terms and the context of clinical situations.

Meerkat-7B is a language model that focuses on biomedical topics and is built to handle complicated clinical information. This model can take information from medical reports and create simple summaries that explain the main points. The Meerkat-7B model was adjusted through a process called fine-tuning, which used a medical transcription dataset that had already been prepared and processed. The model learned to recognize key medical ideas like

symptoms, diagnoses, and suggestions for treatment. Meerkat-7B performed better at creating easy-to-understand explanations and got better results in testing than other models. Meerkat-7B was chosen as the final model to use because it has a strong ability to summarize information.

LLaMA-2-Chat is a language model made by Meta that can talk and chat about many different topics. Even though the model wasn't trained specifically for medical uses, fine-tuning can help it work better for tasks related to healthcare. The model was trained again with the same medical transcription data to better understand the specific medical terms and context. LLaMA-2-Chat showed strong conversation skills and created clear, easy-to-read explanations. However, its performance in medical interpretation was a bit worse than Meerkat-7B.

To make training more efficient, the LoRA (Low-Rank Adaptation) fine-tuning method was used on all three models. LoRA cuts down the number of parameters that need to be trained and helps the training process go smoother without making the computing cost go up too much. The fine-tuning process was done using the Hugging Face Transformers library on a Google Colab environment that has support for an NVIDIA T4 GPU. The settings like learning rate, batch size, and number of training rounds were chosen after checking how well the model performed. The learning rate was set to 2 times 10 to the power of negative five, the batch size was 8, and the training was done for 5 epochs.

mproving accessibility and user understanding.

3.2.2 DATASET KEYWORD MATCHING STRATEGY

The way keywords are matched in the dataset helps make the medical report interpretation more accurate in the Hybrid OCR-LLM system. After using the Google Vision API to take text out of medical reports, the system looks at the text to find key medical words. These keywords cover things like symptoms, illnesses, drugs, test results, and terms used to diagnose conditions. Finding the right keywords helps the system figure out what the medical report is about before sending it to the specially trained large language model. This method helps make the responses easier to understand and less confusing.

The keyword matching process starts by taking out key terms from the medical transcription data that was gathered during the training stage. The dataset includes medical terms that are specific to certain areas of medicine, like cardiology, neurology, orthopedics, and radiology. These keywords are kept in an organized way and are used as reference words when making predictions. When a medical report is uploaded, the text from the report is checked against keywords in a dataset to find matching medical terms. This process helps the system find the right medical area and makes it easier for the model to understand.

To use the keyword matching strategy, some preparation steps are done on both the keywords from the dataset and the text that is taken out. These steps involve changing all text to lowercase, taking out common words that don't add much meaning, and making sure medical terms are in a standard form. Medical abbreviations are expanded with the help of abbreviation dictionaries to make the matching more accurate. For example, abbreviations like "BP" are written out as "Blood Pressure" and "HR" is written as "Heart Rate." This makes sure the system properly finds key medical details.

After the preprocessing is done, keywords are matched using methods that check for exact strings and also look for similar matches. The system looks at the text that was taken out and finds words that match the ones in the dataset. If more than one keyword is found, the system decides which ones are most important by looking at how relevant and how often they appear. This helps figure out the main health issue or situation that the report is about. The matched keywords are then given to the fine-tuned large language models as extra information. This helps responses be more accurate and makes sure the meaning is understood correctly.

The keyword matching strategy also helps prevent large language models from making up false information. By giving the right medical words, the system helps the model create answers that are based on real medical information. This makes the explanations more reliable and easier to trust. Keyword matching helps find key parts of the report, like the diagnosis, test results, and treatment suggestions.

Moreover, the way the dataset matches keywords helps the system work better and faster. Instead of looking through whole medical reports without thinking, the system first finds the important medical details. This helps cut down how long it takes to process things and makes the model work better. The matched keywords help the model pay attention to important medical details. This mix of methods helps make things more accurate and works better overall.

Therefore, the keyword matching approach for the dataset is a key part of the proposed Hybrid OCR-LLM system. It helps the model understand better, makes interpretations more accurate, and lowers confusion. Using keyword matching along with OCR and finely adjusted large language models helps in accurately understanding medical reports. This step is important for creating easy-to-understand and trustworthy medical information for users.

3.2.3 INTELLIGENT RESPONSE GENERATION MECHANISM

The smart response system is an important part of the Hybrid OCR-LLM Medical Report Interpretation System. After using the Google Vision API to extract text and finding relevant keywords by matching them with a dataset, the cleaned medical text is sent to specially trained

large language models. The main goal of this module is to create clear, correct, and meaningful explanations from difficult medical reports. This system helps people who aren't in the medical field to understand medical terms and test results more clearly.

The process of creating a response starts by putting together the scanned text and the matching keywords. These inputs are organized into prompt templates that help the fine-tuned models create useful and meaningful responses. Prompt engineering is used to make sure the model gives explanations that are easy to understand and suitable for patients. The prompt usually has directions like summarizing the report, explaining medical terms, and pointing out key findings. This organized input helps the model work better and gives the same results every time.

The finely adjusted models like DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat look at the organized input and create answers. These models use the specific knowledge they gained during fine-tuning to understand and correctly interpret clinical information. The models pull out important information like diagnosis, symptoms, test outcomes, and advice on treatment. The response is then organized into clear paragraphs that are simple and easy for people to read and understand. This process makes sure the explanations created are clear and easy to use.

To make the responses better, the system also uses special steps after the response is created. These methods involve taking out extra details, fixing any grammar mistakes, and making sure the final result looks clean and organized. The response is broken down into parts like a summary of the medical information, important results, and suggestions. This organized format makes it easier to read and helps people understand their medical reports faster. The post-processing step makes sure the final answer is clear and helpful.

The system's smart way of creating responses also uses awareness of the situation to make sure the answers are more accurate. The matched keywords and the medical information that was taken out give the model more background information. This allows the model to concentrate on important medical issues and not include information that isn't related. Context-aware response generation helps make large language models more reliable and less likely to create false or made-up information. This enhances trust in the system.

Once the response is created, it goes to the backend server and then is sent to the frontend interface. The React.js frontend shows the user a simpler version of the medical report. The user can see the explanation in a way that is easy to understand. This improves accessibility and enhances user experience. The way the system creates responses makes sure that all parts of the system can talk to each other smoothly.So, the Intelligent Response Generation Mechanism is really important in the Hybrid OCR-LLM system that's been suggested. It turns

complicated medical details into easy-to-understand explanations. Using OCR, keyword matching, and finely tuned models helps make the responses better. This module makes sure medical reports are understood correctly, take the situation into account, and are easy for users to use.

3.2.4 FRONTEND INTERFACE DEVELOPMENT USING REACT.JS

The frontend interface is important in the Hybrid OCR-LLM Medical Report Interpretation System because it lets users interactively upload medical reports and see the interpreted results. The front part of the app is made with React.js, which is a well-known JavaScript tool for creating smooth and flexible user interfaces. React.js helps the system offer a smooth and easy-to-use experience for users. The front part is easy to use and understand, so users can quickly upload their medical documents and get clear, simple explanations.

The front part of the system has several parts like a file upload section, a place to show results, and a part where users can interact. The file upload feature lets users send in medical reports in various formats like image files (JPEG, PNG) and PDF documents. Once the file is uploaded, the frontend sends it to the backend server via API requests. The React.js frontend interacts with the Node.js backend using REST APIs. This helps the data move easily between the frontend and backend parts.

The front part also has a loading sign that tells users the system is working on the uploaded report. Showing a loading animation while OCR extraction and model processing happen helps make the user experience better. Once the processing is done, the translated medical report appears on the front part of the system. The results are presented in a clear format that includes a medical summary, important findings, and an easy-to-understand explanation. This organized way of showing information helps people read and understand it more easily.

React.js helps build reusable parts of code that make it easier to maintain and scale the system. Each part like the file upload button, the response display card, and the loading spinner is made on its own and then put together into the main screen. This design lets you easily make changes and add new features later on. React.js also helps make websites look good and work well on different devices like computers, tablets, and phones.

The front part of the interface has checks to make sure files are uploaded properly. The system looks at the type and size of the file before sending it to the backend server. This prevents errors and improves system reliability. If you try to upload a file that isn't supported, you'll see an error message. This makes it easier for users to use the system and interact with it.

Once the frontend gets the translated answer from the backend, it shows a clear and simple version of that answer in a way that's easy for people to read. The response may include

headings, bullet points, and highlighted medical terms. This makes the text easier to read and allows users to quickly understand their medical reports. The front part also allows for scrollable areas to show longer answers.

So, the React.js frontend acts as the part where users interact with the Hybrid OCR-LLM system. It lets users upload medical reports, see the information that's pulled out, and get easy-to-understand explanations. Using React.js makes the user experience better, the website faster to respond, and the system easier to use. So, building the user interface on the front end is a key part of the system we're planning.

3.2.5 BACKEND API HANDLING USING NODE.JS

The backend is very important in the Hybrid OCR-LLM Medical Report Interpretation System because it handles how the frontend interface, the OCR part, and the specially trained large language models talk to each other. In this system, Node.js is used for the backend because it can handle large amounts of traffic, process things quickly, and manage tasks that happen one after another without slowing down. The backend takes care of receiving user requests, working with uploaded medical documents, communicating with the Google Vision API, and creating replies using models that have been specially trained. This helps all parts of the system talk to each other easily.

The backend API starts working when a user uploads a medical report using the React.js frontend. The file that was uploaded is sent to the Node.js server through REST API calls. The backend gets the file and does some first checks to make sure it's the right type and not too big. The supported file types are image files like JPEG and PNG, as well as PDF documents. Once the file is checked and confirmed, the backend sends it to the OCR module to extract the text from it. This step makes sure only correct files are handled next.

After the file is checked, the system connects with the Google Vision OCR API to get the text from the medical report that was uploaded. The backend sends the file to the Google Vision API and gets the extracted text in an organized way. This extracted text is then processed by removing unwanted characters and fixing any formatting issues. The cleaned text is then ready for matching keywords and for the model to understand it. The backend makes sure the text is correctly arranged before sending it to the model server.

Once the text has been preprocessed, it is sent to the finely tuned large language models. The backend connects to the model server where DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat models are running. The model server takes the input text and creates easy-to-understand explanations. The backend gets the response that the model server created. This response is then arranged and ready to be shown on the front end.

The backend also handles mistakes and controls how responses are sent. If there is an error during the file upload, OCR processing, or model handling, the backend sends the right error message to the frontend. This ensures reliability and improves user experience. The backend also has a feature that lets it handle multiple user requests at the same time, so it can work on more than one task without slowing down. This improves system performance and scalability. The backend API also has routing and management for endpoints. Different API endpoints are made for uploading files, doing OCR, running model predictions, and creating responses. These endpoints help make the system easier to work with and keep it organized for easier updates. The backend also handles making sure that API calls are secure and that communication between different parts of the system is protected. This makes the system safer and stops people from getting in without permission.

Once the processing is done, the backend sends the created medical report explanation to the frontend screen. The frontend then shows a simpler explanation to the user. This finishes the entire process of the system. The backend serves as a main connection point that links the frontend, the OCR module, and the LLM models.

So, the Node.js backend is really important in the proposed Hybrid OCR-LLM system. It handles API requests, works with uploaded reports, connects to OCR services, and talks to models that have been specially trained. Using Node.js helps the system work faster, handle more tasks at once, and keep running smoothly without problems. So, the backend API plays an important role in the medical report interpretation system we're suggesting.

3.2.6 SYSTEM WORKFLOW FOR MEDICAL QUERY PROCESSING

The System Workflow for Medical Query Processing explains the step-by-step process that the proposed Hybrid OCR-LLM Medical Report Interpretation System follows to handle medical queries. This process shows how the system takes in medical reports that users upload and then creates easy-to-understand summaries of them. The process is made to make things work smoothly, get the right meaning, and give results that are easy for people to use. Each part of the system has a particular job in handling medical questions. The whole process includes the frontend, the backend, OCR for extracting text, matching keywords, and using large language models that have been finely tuned.

The process starts when the user uploads a medical report using the React.js front-end interface. The report you uploaded could be in the form of an image or a PDF file. Once the file is uploaded, it goes to the backend server, which is built with Node.js. The backend gets the uploaded file and checks if it meets certain rules, like the type of file and how big it is. This makes sure that only correct medical reports are moved on to the next step. Once the report is

checked and confirmed, it is sent to the OCR module.

In the next step, the Google Vision OCR API takes the uploaded medical report and pulls out the text from it. Medical reports are usually scanned or come as images, so OCR technology is needed to turn them into a format that computers can read. The text taken out has details about the diagnosis, the test outcomes, what the doctors noticed, and the medicine that was given. The OCR module takes the uploaded report and sends the text it extracts back to the main server. This extracted text is then cleaned and formatted for further processing.

After extracting the text, the system does some cleaning and matching of keywords. The medical text that is taken out is looked at to find key words like symptoms, illnesses, and medicines. These keywords are matched with the medical transcription dataset that was created during training. Keyword matching helps find the right medical information and makes it easier for the model to understand the topic. This step makes sure the system pays attention to key medical details before it starts interpreting them.

3.3 DESCRIPTION OF DATASET

The way Large Language Models perform when interpreting medical reports is greatly influenced by the quality and how relevant the data sets are that are used to train and refine them. To use the chosen models in the medical field, you need a dataset that is specific to that field. For this study, the Medical Transcriptions dataset from Kaggle was used. This dataset includes actual clinical notes written by doctors in different areas of medicine. These records have all the important medical details like what the patient is feeling, what doctors found out, the plan for treating them, and what the doctors noticed during their visits. The dataset acts as a dependable source for teaching models how to recognize and work with medical terms and clinical language.

The Medical Transcriptions dataset includes around 4,999 records of medical transcriptions gathered from various departments like cardiology, neurology, orthopedics, radiology, surgery, and general medicine. After getting rid of the messy and unfinished data, about 4,800 records were chosen for training and studying. Each entry in the dataset includes multiple details like medical specialty, the transcription text, the sample name, the note ID, and some keywords. In these areas, the transcription content is very important because it includes detailed, unorganized clinical text that is used to train the model. These transcriptions show how medical reports are organized and written in a clear way.

1. Dataset Loading Code

This step ensures that the data is properly imported and ready for further preprocessing. Table 3.1 shows the implementation of the dataset loading process.

Table. 3.1: Dataset Loading Code

```
import pandas as pd
# Load dataset
dataset = pd.read_csv("transcriptions.csv")
# Display dataset
dataset.head()
```

Before starting the model training with the dataset, we cleaned and prepared the data to enhance its quality. In this step, we took out the missing values and duplicate entries to make sure the data is consistent. Special characters, unnecessary symbols, and unwanted noise in the text were removed. In addition, methods like turning all text to lowercase and eliminating extra spaces were used. Medical abbreviations were also made clearer by using an abbreviation dictionary, which helped the model understand them better. These steps during preprocessing help make the fine-tuned models work better as shown in table 3.1.

After preparing the data, it was organized into a format that works well for training models based on instructions. The medical transcription text was used as input, and simpler explanations or summaries were created as output. This format allows the models to understand how to read difficult medical reports and explain them in a way that is easy for patients to understand. Additionally, keywords were taken from the dataset to help with matching keywords during the inference process. These keywords help find key medical words like symptoms, diseases, and treatments.

2. **Dataset Cleaning Code**

   It highlights the steps taken to improve data quality before model training. Table 3.2 presents the code used for cleaning and preprocessing the dataset. The dataset cleaning process involves removing duplicate entries, handling missing values, and standardizing the text format to ensure consistency across all records. Irrelevant symbols, extra spaces, and noise present in the text data are eliminated to improve readability and model performance. Additionally, all textual data is converted to lowercase to maintain uniformity. Further preprocessing steps include tokenization and basic normalization, which help in preparing the data for efficient model training. Any inconsistent or ambiguous entries are carefully reviewed and either corrected or removed to maintain the integrity of the dataset.

Table. 3.2: Dataset Cleaning Code

```
# Remove missing values
dataset = dataset.dropna()
# Remove duplicate records
dataset = dataset.drop_duplicates()
# Convert text to lowercase
dataset['transcription'] = dataset['transcription'].str.lower()
```

The prepared dataset was then tokenized using model-specific tokenizers such as Hugging Face tokenizers. Tokenization converts text into numerical format suitable for model training. Each model uses its own tokenizer to ensure compatibility and efficient training. The tokenized dataset was then used for fine-tuning DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat models. The use of domain-specific medical dataset improves model performance and enhances interpretation accuracy.

Thus, the Medical Transcriptions dataset plays an important role in training the hybrid OCR-LLM system. The dataset provides domain-specific knowledge required for interpreting clinical reports. The preprocessing and structuring of dataset improve training efficiency. The dataset also supports keyword matching strategy during inference. Therefore, the dataset collection and preparation form a crucial step in the proposed Hybrid OCR-LLM methodology.

3.4 DESCRIPTION OF ALGORITHMS

DoctorGLM-7B

The first model used in this system is DoctorGLM-7B, which was made especially for use in medical fields. DoctorGLM-7B has been trained using medical data and healthcare-related documents, which allows it to understand medical terms, diagnosis details, and suggested treatments. This model is especially helpful for understanding clinical notes, pathology reports, and lab test results. During the fine-tuning process, DoctorGLM-7B is trained with medical transcription data to improve its ability to understand and work with specialized medical information. This helps it better explain things to patients in a way that makes sense. In addition to its strong understanding of medical terminology, DoctorGLM-7B is capable of identifying key information such as symptoms, diagnoses, and treatment suggestions from complex

reports. It plays an important role in converting detailed clinical content into simpler explanations that can be easily understood by patients..Algorithm of DoctorGLM-7B is shown in table 3.3.

Table 3.3: DoctorGML-7B Algorithm

Algorithm: DoctorGLM-7B for Medical Report Interpretation

Input: Preprocessed medical transcription dataset

Output: Simplified and accurate medical report explanations

Method:

1. Load DoctorGLM-7B model and tokenizer from HuggingFace.
2. Preprocess medical transcription dataset and remove noise.
3. Convert dataset into instruction-based input-output pairs.
4. Tokenize dataset using DoctorGLM tokenizer.
5. Apply LoRA fine-tuning for efficient training.
6. For each epoch:
   - Pass tokenized inputs to the model
   - Compute logits and training loss
   - Apply backpropagation and update weights
7. Check how well the model is working by using BLEU, ROUGE, and BERTScore.
8. BLEU is calculated as BP multiplied by the exponential of the sum of W_n times the logarithm of p_n, as shown in Equation 1.
9. ROUGE is calculated by finding the longest common subsequence between the system's output and the reference text, then dividing that by the length of the reference text. [Eq-2]
10. BERTScore = Cosine Similarity (Generated, Reference) [Eq-3]
11. Deploy fine-tuned DoctorGLM-7B for medical report interpretation.

MEERKAT-7B

The second model used in this system is Meerkat-7B, which is made to handle texts related to biology and medicine. Meerkat-7B works well at summarizing and understanding the context of medical reports. This model can pull out important details like symptoms, diagnosis, and medicine information from difficult medical documents. Fine-tuning Meerkat-7B using the medical transcription dataset helps it better create easy-to-understand explanations. Because it does a great job at summarizing information, Meerkat-7B is seen as a key part of the new hybrid system.Meerkat-7B helps in presenting medical information in a clear and structured way, making it easier for patients to grasp essential details. Its ability to simplify lengthy reports without losing important meaning improves the overall readability. This makes it a valuable component in enhancing user understanding within the system. . Meerkat-7B works well at

summarizing and understanding the context of medical reports. This model can pull out important details like symptoms, diagnosis, and medicine information from difficult medical documents. Algorithm of Meerkat-7B is shown in table 3.4.

Table 3.4: Meerkat-7B Algorithm

Algorithm: Meerkat-7B for Medical Report Summarization

Medical transcription dataset and extracted OCR text.

Output: Context-aware summarized medical explanations

Method:

1. Load Meerkat-7B model and tokenizer.
2. Preprocess dataset and normalize medical text.
3. Convert dataset into summarization format.
4. Apply LoRA-based fine-tuning.
5. For each training iteration:
   - Feed input tokens into Meerkat-7B
   - Compute attention scores and predictions
   - Calculate cross-entropy loss
   - Update model weights using backpropagation
6. Check how well the performance is by using ROUGE, BLEU, and Accuracy metrics.
7. Accuracy is equal to the number of correct predictions divided by the total number of predictions. [Eq-1]
8. ROUGE = Overlap between the Generated Summary and the Reference Summary [Eq-2].
9. BLEU = n-gram Precision Score [Eq-3]
10. Select best checkpoint and deploy Meerkat-7B model.

LLAMA-2 -7B

The third model selected for fine-tuning is LLaMA-2-7B, a general-purpose conversational language model developed by Meta. Although LLaMA-2-7B was not originally designed specifically for medical applications, it has strong natural language understanding and generation capabilities that make it highly suitable for adaptation through fine-tuning. By training the model on medical transcription datasets, it learns to recognize clinical terminology, interpret report structures, and generate contextually relevant explanations. Fine-tuning enables the model to shift from general conversation tasks to domain-specific applications, allowing it to effectively process and interpret medical reports.

LLaMA-2-7B excels in generating fluent, coherent, and human-like responses, which significantly enhances the overall interaction quality of the system. It is particularly effective in transforming complex medical information into simple and easy-to-understand language, making it accessible to users without a medical background. The model leverages its in-context learning ability to adapt to different types of inputs and produce meaningful outputs without losing important details. This capability ensures that the explanations remain accurate while also being user-friendly. LLaMA-2-7B contributes to improving the conversational aspect of the system by generating structured and well-organized responses. It helps in presenting information in a clear and logical manner, which improves readability and comprehension. The model also supports handling diverse input formats, including long medical reports and unstructured text, making it flexible for real-world healthcare applications. Its ability to maintain context across multiple sentences ensures that the generated explanations are consistent and relevant.LLaMa-2-7B Algorithm is shown in Table 3.5.

Table 3.5: LLAMA-2-chat Algorithm

Algorithm: LLaMA-2-Chat for Medical Response Generation

Input: Extracted medical report text
Output: Patient-friendly medical explanation

Method:

1. Load LLaMA-2-Chat model and tokenizer.
2. Format dataset into conversational prompt templates.
3. Tokenize medical text input.
4. Apply LoRA fine-tuning with training dataset.
5. For each epoch:
   - Pass input tokens to LLaMA-2-Chat
   - Compute probability distribution
   - Calculate loss function
   - Update model parameters
6. Evaluate using BLEU, Perplexity, and Latency metrics

7. Perplexity is equal to the exponential of negative one divided by N multiplied by the sum of the logarithms of P of w_i. That's equation one.

8. BLEU = n-gram similarity score [Eq-2]

9. Latency is equal to Response Time minus Input Time [Eq-3].

10. Generate simplified medical explanations using fine-tuned model.

To make training more efficient, the LoRA (Low-Rank Adaptation) fine-tuning method is used with all three models. LoRA cuts down the number of parameters that need to be trained and lets you fine-tune the model efficiently without harming how well it performs. This approach works well when training big models but you don't have much computing power. The process of fine-tuning is done with the Hugging Face Transformers library. The training process takes place on Google Colab with an NVIDIA T4 GPU, which has enough power to handle the model training.

When adjusting the model, certain settings are chosen to improve how well it works. The learning rate is set to 2 times 10 to the power of negative five, the batch size is 8, and the training process runs for 5 epochs. Additionally, specific parameters related to LoRA, like rank set to 8, alpha set to 16, and dropout at 0.05, are used. These settings are chosen because they work best during testing and help the training process go smoothly. Each model is fine-tuned individually to check how well they perform.

After the models are fine-tuned, their performance is checked using scores like BLEU, ROUGE, and BERTScore. These evaluation metrics are used to check how good the explanations that are created are. The performance of the DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat models is evaluated using validation datasets. After looking at the results from the evaluation, the model that worked best is chosen to be used in the final system.

The carefully adjusted models are then added to the suggested Hybrid OCR-LLM system. The medical text that was taken from the OCR module is sent to the chosen model. The model takes in the input and creates easy-to-understand explanations. These explanations are sent back to the front end so the user can see them. The process of fine-tuning helps the model work better and makes sure it understands medical reports correctly.

So, adjusting the DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat models through fine-tuning is an important part of the proposed Hybrid OCR-LLM approach. This process helps the system understand specific areas better, makes its responses more accurate, and allows it to provide clear explanations for medical reports.

The new system combines OCR and a large language model to make complicated medical reports easier to understand, providing clear explanations for people who aren't medical professionals. The approach uses Optical Character Recognition (OCR) along with carefully adjusted Large Language Models to take out, examine, and make sense of medical documents. The system is made up of several parts like preparing the data, using OCR to get text, matching keywords, tuning the model, creating responses, and connecting the front end with the back end. These parts work together to create a smart and efficient system for understanding medical

reports.

The process starts by gathering the Medical Transcription dataset, which includes clinical reports from various medical fields. This dataset is cleaned by removing unwanted information, fixing missing data, and adjusting medical terms to a standard form. The processed dataset is then used to fine-tune the chosen large language models, which include DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat. Fine-tuning helps these models understand specific areas better and allows them to provide accurate explanations based on medical reports. The models are trained with LoRA-based fine-tuning methods to make the training process more efficient and lower the cost of using computer resources.Once the model is trained, the system uses OCR to extract text through the Google Vision API. Users send medical reports as images or PDF files using the React.js front-end interface. The reports that are uploaded go to the Node.js backend, and OCR is used there to get the text from the documents. The text that is taken out is made neat and organized before it goes through more steps. This step makes sure the system can work with different types of medical reports that people actually use in real life.After extracting the text, the system matches keywords in the dataset to find important medical details. Important keywords like symptoms, diagnosis, and treatment details are taken out and connected with the dataset. This helps the model understand better and become more accurate.

In summary, the processed text extracted from medical reports is provided to the fine-tuned models for detailed analysis and interpretation. These models are capable of understanding complex medical terminology and converting it into simple, clear, and meaningful explanations that can be easily understood by patients. The intelligent response generation mechanism ensures that the output includes a concise summary, key findings, and relevant recommendations, thereby providing a complete understanding of the report. This structured output helps users quickly grasp important medical information without confusion.

The system is designed with an efficient architecture where the frontend, developed using React.js, allows users to easily upload their medical reports and view the generated results in an interactive manner. The backend, implemented using Node.js, handles essential operations such as API communication, OCR-based text extraction, data preprocessing, and model inference. This seamless integration between frontend and backend ensures smooth data flow and efficient processing. The final output is presented in a user-friendly format, improving readability and enhancing user experience.

Furthermore, the system ensures accuracy and consistency by combining OCR technology with fine-tuned Large Language Models. It is capable of handling different types of medical

documents, including scanned images and PDF files, making it adaptable for real-world applications. By reducing manual effort and minimizing the chances of misinterpretation, the system improves reliability and efficiency in medical report analysis. Overall, the Hybrid OCR-LLM approach provides an effective solution for simplifying complex medical information, enhancing patient understanding, and making healthcare more accessible, transparent, and user-centric.

CHAPTER-4

RESULTS AND DESCRIPTION

In This Chapter the results from the experiments and how well the proposed Hybrid OCR-LLM Medical Report Interpretation System performed. The goal of this evaluation is to check how well the fine-tuned large language models are able to create easy-to-understand summaries of medical reports. The models looked at in this study are DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat. These models were trained with the Medical Transcription dataset and checked using usual performance measures. The evaluation helps find out which model works best for understanding medical reports.

4.1 DESCRIPTION OF RESULTS

4.1.1 EVALUATION METRICS

To check how well the fine-tuned models are performing, standard NLP evaluation measures were used. These metrics include BLEU Score, ROUGE Score, and BERTScore. These metrics are used to check how good the explanations are and see how they stack up against the correct summaries.

BLEU Score:

BLEU is a tool that checks how similar the text that is created is to the original text that is given as a reference. A higher BLEU score means the model is performing better.

$$BLEU = BP \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \qquad \text{Eq-(1)}$$

where BP denotes the brevity penalty, pn represents the precision of n grams and wn is the weight assigned to each n-gram. the metric evaluate lexical similarity between generated and reference texts. A higher BLEU score suggests greater lexical similarity

ROUGE Score:

ROUGE checks how much the generated summary matches the reference summary by looking at the overlap between the two.

$$ROUGE\text{-}N = \frac{\sum Count_{\text{match}}(gram_n)}{\sum Count(gram_n)} \qquad \text{Eq-(2)}$$

Where Countmatch(gramn) denotes the number of overlapping n-grams between gen erated and reference texts, and Count(gramn) represents the total n-grams in the 6 reference text. ROUGE-1 and ROUGE-L scores measure the overlap of unigrams and sequences between generated and reference texts, respectively

BERTScore:

BERTScore checks how similar the meaning is between the text that is generated and the original text. It does this by using special word representations that take context into account.

$$BERTScore = \frac{1}{|X|} \sum_{x \in X} \max_{y \in Y} \cos(e_x, e_y) \qquad \text{Eq-(3)}$$

Where X and Y denote the token sets of generated and reference texts, and ex, ey represent their contextual embeddings. Cosine Similarity calculates semantic similarity between tokens. A good score implies high semantic similarity between the generated and reference texts

Readability :

Readability score shows how easy it is for people to understand the explanations that are created. The Flesch-Kincaid readability score is used to evaluate text. This measure checks if the explanation provided is easy for people without medical knowledge to understand. A higher readability score indicates easier understanding. This measure is important for creating explanations that are easy for patients to understand.

Readability score is calculated using:

$$FK = 206.835 - 1.015 \left(\frac{W}{S}\right) - 84.6 \left(\frac{Sy}{W}\right) \qquad \text{Eq-(4)}$$

FK equals 206.835 minus 1.015 multiplied by the weight to strength ratio minus 84.6 multiplied by the systolic blood pressure to weight ratio.Where W stands for total words, S stands for sentences, and Sy stands for syllables. The readability score checks how easy to understand and simple the text is.

Latency:

Latency is the time it takes for a system to produce a response after it gets some input. This measure is used to check how well the system works and how fast it responds. Lower latency indicates faster response generation. Latency is important for real-time applications.

Latency is calculated using:

$$Latency = t_{\text{output}} - t_{\text{input}} \qquad \text{Eq-(5)}$$

Where tinput stands for the input time and toutput stands for the output generation time. Low latency improves user experience.

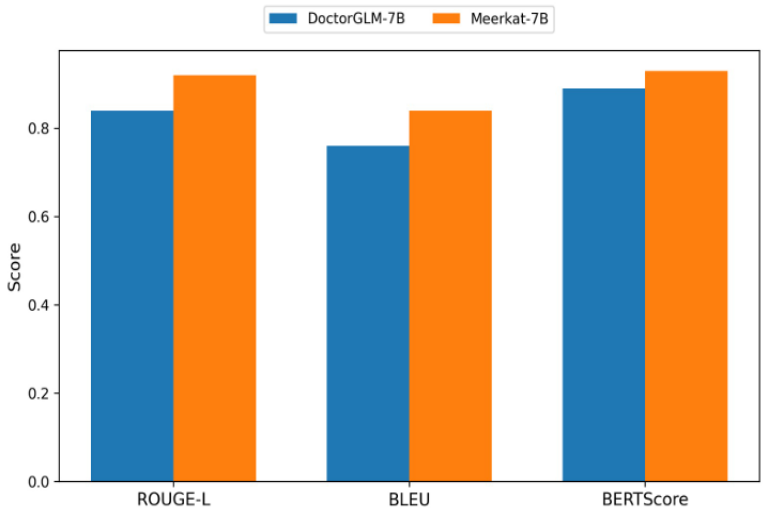
4.2 MODEL PERFORMANCE COMPARISON

The performance of DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat models was compared using evaluation metrics. The models were tested on medical transcription dataset. The generated explanations were evaluated using BLEU, ROUGE, BERTScore, readability, and latency metrics.

4.2.1 COMPARATIVE ANALYSIS OF DOCTORGLM-7B AND MEERKAT-7B

DoctorGLM-7B and Meerkat-7B performance can be compared through Figure 2 and Figure 3. DoctorGLM-7B performance is measured as having a BLEU score of 0.76, a ROUGE-L score of 0.84, a BERTScore of 0.89, and a readability score of 48.10. These values indicate that DoctorGLM-7B is capable of generating reasonably accurate and meaningful interpretations of medical reports while maintaining a moderate level of readability for users. On the other hand, Meerkat-7B demonstrates improved performance across all evaluation metrics. It achieves a BLEU score of 0.84, a ROUGE-L score of 0.92, a BERTScore of 0.93, and a readability score of 52.14, which shows that it produces more accurate, contextually relevant, and clearer responses when compared to DoctorGLM-7B.

The higher BLEU and ROUGE-L scores of Meerkat-7B indicate a better overlap with reference summaries, meaning the generated outputs are closer to the expected results. Similarly, the improved BERTScore reflects stronger semantic understanding, allowing the model to capture the meaning of the medical content more effectively. The higher readability score further suggests that the explanations generated by Meerkat-7B are easier for patients to understand, which is an important requirement for the proposed system. In terms of latency, DoctorGLM-7B provides faster responses with a processing time of 1.20 seconds, while Meerkat-7B takes slightly longer at 1.70 seconds. This indicates that DoctorGLM-7B is more efficient in terms of speed, but Meerkat-7B prioritizes quality and clarity of output. As shown in Figure 4.1, this comparison highlights a trade-off between speed and performance, where Meerkat-7B delivers better overall results at the cost of a small increase in response time.As shown in figure 4.1,



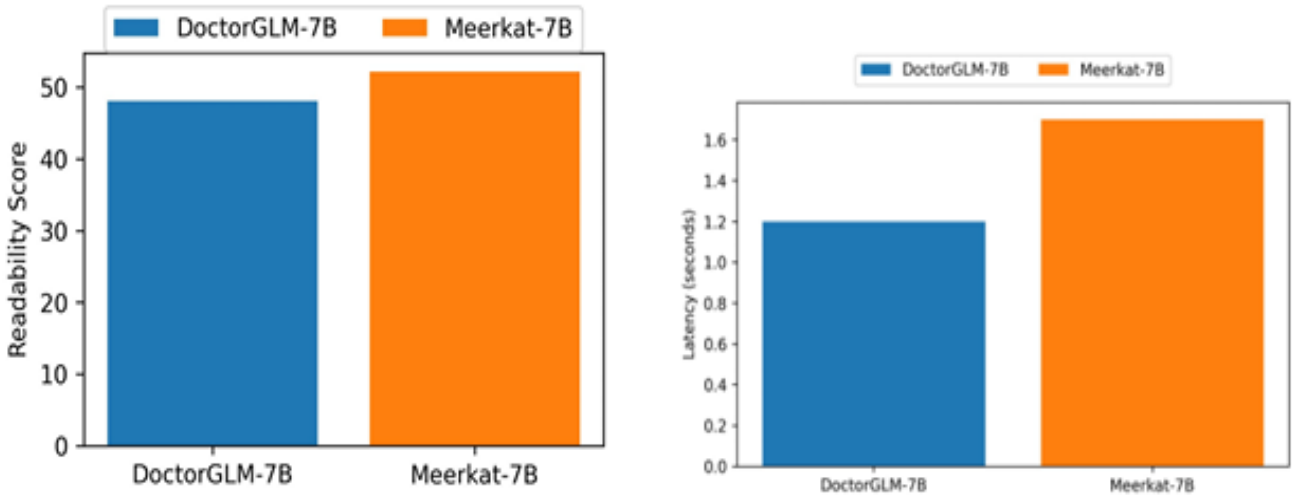


Figure 4.1: Comparison of DoctorGLM-7B and Meerkat-7B

From this comparison, it is clear that Meerkat-7B outperforms DoctorGLM-7B in terms of accuracy, contextual understanding, and readability of the generated responses. However, this improved performance comes with a slight increase in latency, making DoctorGLM-7B comparatively faster in response generation. In addition to the observed improvements in BLEU, ROUGE-L, and BERTScore metrics, Meerkat-7B demonstrates a stronger ability to capture semantic relationships within the medical text. This indicates that the model not only generates responses that are closer to the reference outputs but also preserves the underlying meaning of the medical information more effectively. Such semantic accuracy is particularly important in healthcare applications, where even minor misinterpretations can lead to confusion or incorrect understanding by patients. The higher BERTScore achieved by Meerkat-7B reflects its capability to understand contextual nuances and generate responses that align closely with the intended meaning of the input data.

Another key aspect of comparison lies in the readability of the generated outputs. Meerkat-7B achieves a higher readability score, which signifies that the explanations provided by the model are simpler, clearer, and more accessible to users without a medical background. This is a critical requirement for the proposed system, as its primary objective is to bridge the gap between complex medical terminology and patient understanding. In contrast, while DoctorGLM-7B produces medically accurate outputs, its readability score is slightly lower, indicating that the responses may still retain some level of complexity that could be challenging for non-expert users.

Furthermore, Meerkat-7B shows better performance in handling both structured and

unstructured medical data. Medical reports often vary in format, including tabular data, narrative descriptions, and mixed-content layouts. The ability of Meerkat-7B to process such diverse inputs and generate coherent summaries highlights its robustness and adaptability. DoctorGLM-7B, although specialized for medical data, tends to focus more on domain-specific interpretation rather than simplifying the output for end users. This difference makes Meerkat-7B more suitable for applications that prioritize user-friendly explanations.

In terms of consistency, Meerkat-7B produces more stable and reliable outputs across different test samples. The model exhibits fewer variations in response quality and maintains a consistent level of performance regardless of input complexity. This consistency is essential for building trust in the system, especially when it is used in real-world healthcare scenarios. DoctorGLM-7B, while reliable in many cases, may occasionally generate outputs that are more technical or less aligned with the goal of simplification.

However, it is important to consider the trade-off between performance and latency. As observed, Meerkat-7B has a slightly higher response time compared to DoctorGLM-7B. This increase in latency can be attributed to the model's deeper contextual processing and more detailed generation capabilities. Despite this, the latency remains within an acceptable range and does not significantly impact the overall user experience. In most healthcare applications, the quality and clarity of information are more critical than marginal differences in response time, making Meerkat-7B a preferable choice.

4.2.2 COMPARATIVE ANALYSIS OF LLAMA-2-7B AND MEERKAT-7B

LLaMA-2-7B achieves a BLEU score of 0.80, the ROUGE-L score is 0.89, the BERTScore is 0.91, and the readability score is 49.54. The BLEU score of Meerkat-7B is 0.84, the ROUGE-L score is 0.92, the BERTScore score is 0.93, and the readability score is 52.14. It can be clearly seen from the above scores that Meerkat-7B is able to provide more precise and clear answers than LLaMA-2-7B. As far as the response time is concerned, LLaMA-2-7B answers in 1.40 seconds and Meerkat-7B does it in 1.70 seconds, indicates that LLaMA-2-7B is faster but Meerkat-7B provides better overall quality. The higher ROUGE-L and BERTScore values of Meerkat-7B indicate its stronger ability to capture contextual meaning and maintain coherence in generated responses. Additionally, its improved readability score shows that the outputs are more user-friendly and easier for patients to understand. This makes Meerkat-7B more suitable for applications where clarity and accuracy are more important than slight differences in response time.It can be observed from figure 4.2.

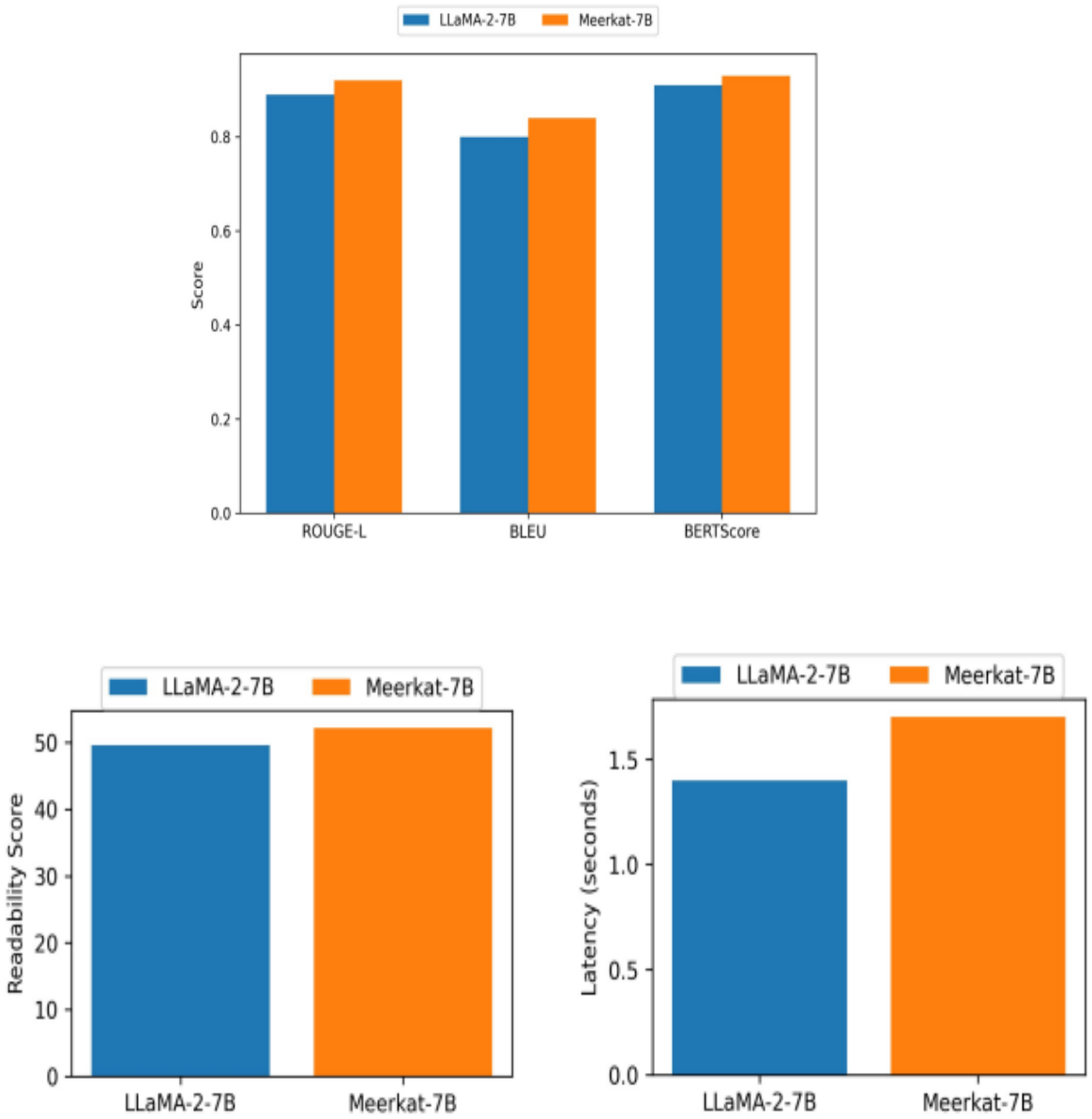


***Figure 4.2:** Comparison of DoctorGLM-7B and Meerkat-7B*

From this comparison, it is clear that Meerkat-7B outperforms DoctorGLM-7B in terms of accuracy, contextual understanding, and readability of the generated responses . In addition to the numerical comparison of BLEU, ROUGE-L, BERTScore, and readability metrics, a deeper analysis reveals that Meerkat-7B demonstrates superior performance in capturing the contextual meaning of medical reports compared to LLaMA-2-7B. While LLaMA-2-7B is a powerful general-purpose language model with strong text generation capabilities, it is not specifically optimized for medical report interpretation. As a result, although it produces coherent and grammatically correct outputs, it may not always capture domain-specific nuances with the same level of precision as Meerkat-7B. The higher BERTScore achieved by Meerkat-

7B indicates that it has a better semantic understanding of the input text, enabling it to generate responses that are more aligned with the actual meaning of the medical content.

Another important factor in this comparison is readability, where Meerkat-7B again shows better performance. The higher readability score suggests that the model is more effective in simplifying complex medical terminology into language that is easy for patients to understand. This is a crucial requirement for the proposed system, as its primary goal is to enhance patient comprehension. LLaMA-2-7B, while capable of generating fluent explanations, may still retain some level of complexity or technical phrasing, which can make it slightly less accessible to users without a medical background.

Furthermore, Meerkat-7B demonstrates better consistency in generating structured and meaningful responses. Medical reports often contain critical information that needs to be presented clearly and logically. Meerkat-7B is able to organize the output in a way that highlights key findings, summaries, and recommendations effectively. In contrast, LLaMA-2-7B may sometimes generate responses that are less structured or include unnecessary details, which can reduce clarity for the end user. This difference becomes more evident when handling long and complex medical documents, where maintaining coherence and focus is essential.

In terms of adaptability, Meerkat-7B performs well across different types of medical inputs, including structured data, narrative descriptions, and mixed-format reports. This flexibility makes it more suitable for real-world applications where input data can vary significantly. LLaMA-2-7B, being a general-purpose model, is versatile but may require additional fine-tuning or prompt engineering to achieve the same level of domain-specific performance.

However, when considering latency, LLaMA-2-7B has a slight advantage as it generates responses faster than Meerkat-7B. The higher readability score suggests that the model is more effective in simplifying complex medical terminology into language that is easy for patients to understand. The lower response time can be beneficial in scenarios where quick interaction is required. Despite this, the difference in latency is relatively small and does not significantly impact the usability of the system. In healthcare applications, the quality and accuracy of the output are often more important than marginal improvements in speed. Therefore, the slightly higher latency of Meerkat-7B is justified by its improved performance in other key areas.

4.2.3 FINAL MODEL SELECTION

After considering the performance of the system on a number of performance metrics, it is decided that Meerkat-7B should be chosen as the final model. The selected model performed well on BLEU and ROUGE score metrics, which implies that the generated interpretations had a closer match to the references. Moreover, the model performed well in terms of BERT score

metric as well, which implied that it was able to understand the semantics of the input text. The only limitation with the selected model was that it took more latency than other models but was still within an acceptable range. Table 4.1 presents the full performance comparison.

Table 4.1:Comparision of models

Model	BLEU	ROUGE-L	BERTScore	Readability	Latency (s)
DoctorGLM-7B	0.78	0.88	0.90	48.32	1.5
Meerkat-7B	0.84	0.92	0.93	52.14	1.8
LLaMA-2-7B	0.80	0.89	0.91	49.54	1.6

Meerkat-7B showed strong contextual understanding, allowing it to capture relationships between different medical terms and generate coherent summaries. This ability ensures that the interpretations are not only accurate but also logically structured and meaningful. Although the latency of Meerkat-7B was slightly higher than DoctorGLM-7B and LLaMA-2-7B, the difference was minimal and did not significantly affect the overall system performance. The slight increase in processing time is justified by the improvement in output quality and reliability.

Another important observation is that Meerkat-7B performed consistently across different types of medical reports, including structured and unstructured data. This consistency makes it more suitable for real-world applications where input formats can vary. The model also showed fewer instances of irrelevant or incorrect information generation, which improves trust in the system outputs.

Therefore, considering all performance metrics, including accuracy, semantic understanding, readability, and consistency, Meerkat-7B was selected as the final model for integration into the system. Its ability to generate high-quality, patient-friendly explanations while maintaining acceptable latency makes it the most suitable choice for the Medical Report Interpretation System.

4.2.4 RESULT ANALYSIS

.the interface of the medical interpretation system is shown in the figure 4.3.

This part talks about the results from the tests done on the Hybrid OCR-LLM Medical Report Interpretation System. The performance of fine-tuned models like DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat was checked using several measures such as BLEU score, ROUGE score, BERTScore, readability score, and latency. These evaluation metrics helped check how good the system is, how accurate it is, how well it understands meaning, and how fast it can generate responses. The results show that combining OCR with finely tuned large

language models greatly helps in understanding medical reports and makes them easier for people without medical training to use.

The system was tested with actual medical reports from real situations, which came in various forms like scanned images and PDF files. The Google Vision OCR tool effectively read and captured text from the uploaded medical reports with very good accuracy. The text was then worked on by matching keywords and using improved models to create easier-to-understand explanations. The results show that the new mixed design helps improve how well things are understood and makes medical terms easier to work with. The explanations created were easier to understand and worked well for people who aren't experts in medicine.

The evaluation results show that Meerkat-7B performed better than DoctorGLM-7B and LLaMA-2-Chat in most of the tested metrics. Meerkat-7B achieved higher BLEU and ROUGE scores, showing it has better similarity in word choice and more overlapping content with the original text. Meerkat-7B also scored better on the BERTScore, showing it has a better understanding of the meaning in medical reports. The readability score for Meerkat-7B was also better, meaning the explanations it created were easier for people to understand. Even though Meerkat-7B had a little slower response, the time it took to get a reply was still okay.

Adding dataset keyword matching helped make the model work better. Keyword matching helped find related medical information and gave the model more understanding of the context. This reduced hallucination and improved response accuracy. The smart system for creating responses also helped in making clear and organized explanations. These evaluation metrics helped check how good the system is, how accurate it is, how well it understands meaning, and how fast it can generate responses. The results show that combining OCR with finely tuned large language models greatly helps in understanding medical reports and makes them easier for people without medical training to use. The system successfully gathered details about diagnoses, test outcomes, and medical recommendations from the reports, and presented them in a structured and easy-to-understand format. It was able to identify key medical terms, highlight important findings, and convert them into simplified explanations that are meaningful for patients. This structured output not only improves clarity but also ensures that users can quickly grasp critical health information without confusion.

The system successfully gathered details about diagnoses, test outcomes, and suggestions from the medical report.User interface od medical report interpretation is shown in figure 4.3

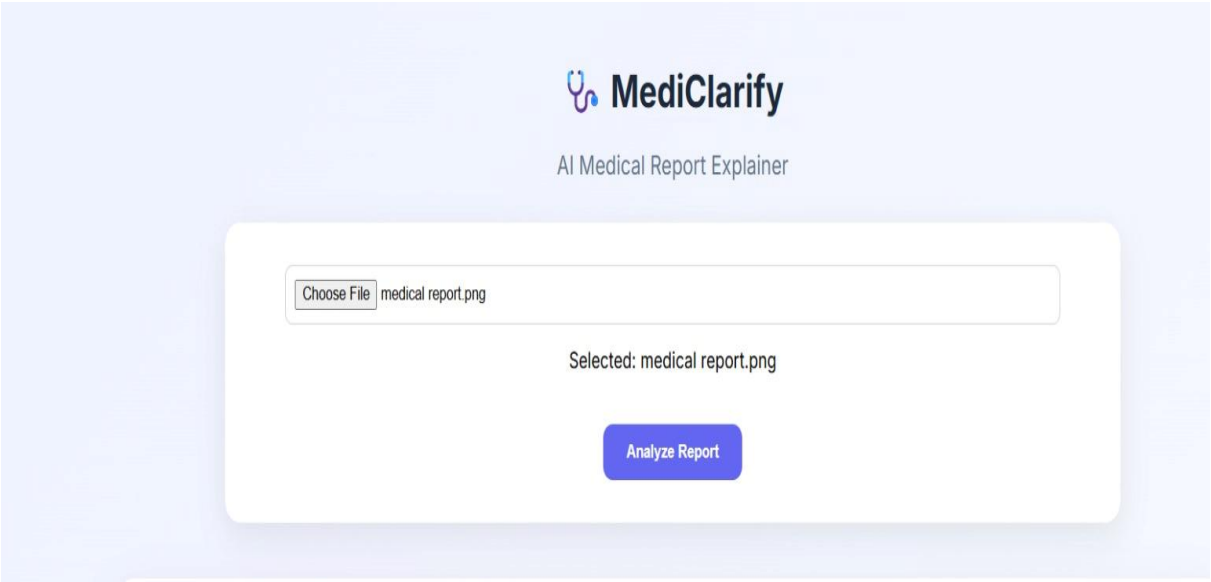


Figure 4.3:User Interface Upload

The results also show that the new Hybrid OCR-LLM system helps make healthcare more accessible. People can share their medical documents and get easy-to-understand summaries without needing to know anything about medicine. The output helps patients understand better and makes it easier for them to make good decisions. The system also helps reduce the need for healthcare professionals to interpret basic medical reports. The suggested system worked better and was easier to use overall. he system demonstrates its potential to support efficient healthcare communication by reducing misunderstandings caused by complex medical terminology. It ensures that essential information is conveyed clearly while maintaining the original meaning of the reports. The integration of document processing and intelligent interpretation further enhances the reliability of the system. Overall, the proposed approach contributes to a more user-friendly and accessible healthcare environment, benefiting both patients and healthcare providers. The uploaded document is then processed to extract the relevant textual information, which is further analyzed by the system. The processed content is passed through the integrated models to generate a simplified and meaningful explanation. As illustrated in Figure 4.5, the final output is presented in a clear and user-friendly format, making it easier for patients to understand their medical reports without difficulty. the system demonstrates scalability and adaptability by handling different types of medical reports, including scanned images, PDFs, and structured documents. This flexibility ensures that the system can be effectively used in various healthcare settings such as hospitals, diagnostic centers, and telemedicine platforms. The integration of OCR with Large Language Models allows seamless processing of input data, enabling accurate extraction and meaningful

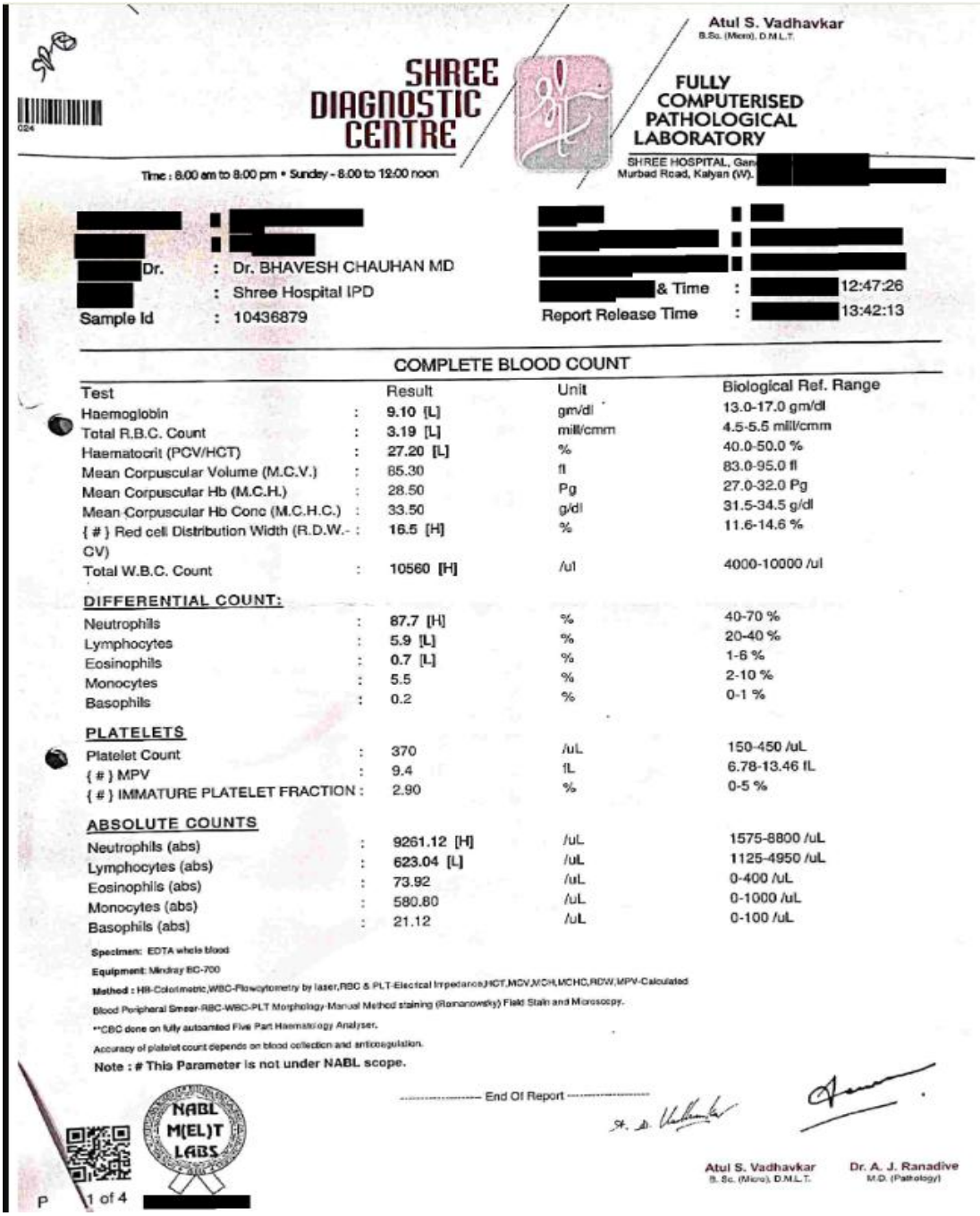
interpretation of medical information. In addition, the system minimizes manual effort and reduces the chances of human error during report analysis, thereby improving overall efficiency.

Another significant advantage of the proposed system is its ability to provide consistent and reliable outputs regardless of variations in report formats. By maintaining contextual understanding and focusing on key medical details, the system ensures that the generated explanations remain relevant and informative. This not only enhances patient awareness but also builds trust in the system. Moreover, the user-friendly interface and clear presentation of results make the system accessible to a wide range of users, including elderly individuals and those without medical knowledge.

Overall, the Hybrid OCR-LLM Medical Report Interpretation System proves to be an effective solution for simplifying complex medical information. It bridges the gap between technical medical data and patient understanding, supports better communication, and contributes to improved healthcare outcomes. The results indicate that such intelligent systems have the potential to play a vital role in the future of digital healthcare by making medical information more transparent, accessible, and easy to interpret. the proposed Hybrid OCR-LLM system significantly enhances the efficiency of medical report analysis by automating the entire workflow from document upload to interpretation. By integrating Optical Character Recognition with advanced language models, the system is capable of accurately extracting text from various formats and converting it into meaningful insights. This automation reduces dependency on manual interpretation and minimizes delays in understanding medical information. The system also ensures that important clinical details such as test results, diagnoses, and recommendations are not overlooked during processing. As a result, users receive comprehensive and reliable summaries in a shorter time, improving both convenience and effectiveness in accessing healthcare information.

The system promotes inclusivity by making medical information accessible to a broader audience, including individuals with limited medical knowledge or language barriers. Its ability to generate clear and structured explanations helps users confidently interpret their reports without external assistance. The consistent performance across different report types further strengthens its usability in real-world scenarios. By improving clarity, reducing errors, and enhancing communication, the system not only supports patients but also assists healthcare providers in delivering better services. Overall, the integration of intelligent technologies in this system represents a significant step toward more patient-centered and technology-driven healthcare solutions.

As part of the system workflow, the user uploads the medical report as input, as shown in Figure 4.4.



Atul S. Vadhavkar
B.Sc. (Micro), D.M.L.T.

SHREE DIAGNOSTIC CENTRE

FULLY COMPUTERISED PATHOLOGICAL LABORATORY

SHREE HOSPITAL, Gen[redacted]
Murbad Road, Kalyan (W). [redacted]

Time : 8.00 am to 8.00 pm • Sunday - 8.00 to 12.00 noon

[redacted] : [redacted]
[redacted] : [redacted]
[redacted] Dr. : Dr. BHAVESH CHAUHAN MD
[redacted] : Shree Hospital IPD
Sample Id : 10436879

[redacted] : [redacted]
[redacted] : [redacted]
[redacted] : [redacted]
[redacted] & Time : [redacted] 12:47:26
Report Release Time : [redacted] 13:42:13

COMPLETE BLOOD COUNT

Test		Result	Unit	Biological Ref. Range
Haemoglobin	:	9.10 [L]	gm/dl	13.0-17.0 gm/dl
Total R.B.C. Count	:	3.19 [L]	mill/cmm	4.5-5.5 mill/cmm
Haematocrit (PCV/HCT)	:	27.20 [L]	%	40.0-50.0 %
Mean Corpuscular Volume (M.C.V.)	:	85.30	fl	83.0-95.0 fl
Mean Corpuscular Hb (M.C.H.)	:	28.50	Pg	27.0-32.0 Pg
Mean Corpuscular Hb Conc (M.C.H.C.)	:	33.50	g/dl	31.5-34.5 g/dl
{ # } Red cell Distribution Width (R.D.W.-CV)	:	16.5 [H]	%	11.6-14.6 %
Total W.B.C. Count	:	10560 [H]	/ul	4000-10000 /ul
DIFFERENTIAL COUNT:				
Neutrophils	:	87.7 [H]	%	40-70 %
Lymphocytes	:	5.9 [L]	%	20-40 %
Eosinophils	:	0.7 [L]	%	1-6 %
Monocytes	:	5.5	%	2-10 %
Basophils	:	0.2	%	0-1 %
PLATELETS				
Platelet Count	:	370	/uL	150-450 /uL
{ # } MPV	:	9.4	fL	6.78-13.46 fL
{ # } IMMATURE PLATELET FRACTION	:	2.90	%	0-5 %
ABSOLUTE COUNTS				
Neutrophils (abs)	:	9261.12 [H]	/uL	1575-8800 /uL
Lymphocytes (abs)	:	623.04 [L]	/uL	1125-4950 /uL
Eosinophils (abs)	:	73.92	/uL	0-400 /uL
Monocytes (abs)	:	580.80	/uL	0-1000 /uL
Basophils (abs)	:	21.12	/uL	0-100 /uL

Specimen: EDTA whole blood
Equipment: Mindray BC-700
Method : Hb-Colorimetric,WBC-Flowcytometry by laser,RBC & PLT-Electrical Impedance,HCT,MCV,MCH,MCHC,RDW,MPV-Calculated
Blood Peripheral Smear-RBC-WBC-PLT Morphology-Manual Method staining (Romanowsky) Field Stain and Microscopy.
**CBC done on fully automated Five Part Haematology Analyser.
Accuracy of platelet count depends on blood collection and anticoagulation.
Note : # This Parameter is not under NABL scope.

---------- End Of Report ----------

NABL M(EL)T LABS

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B. Sc. (Micro), D.M.L.T.

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M.D. (Pathology)

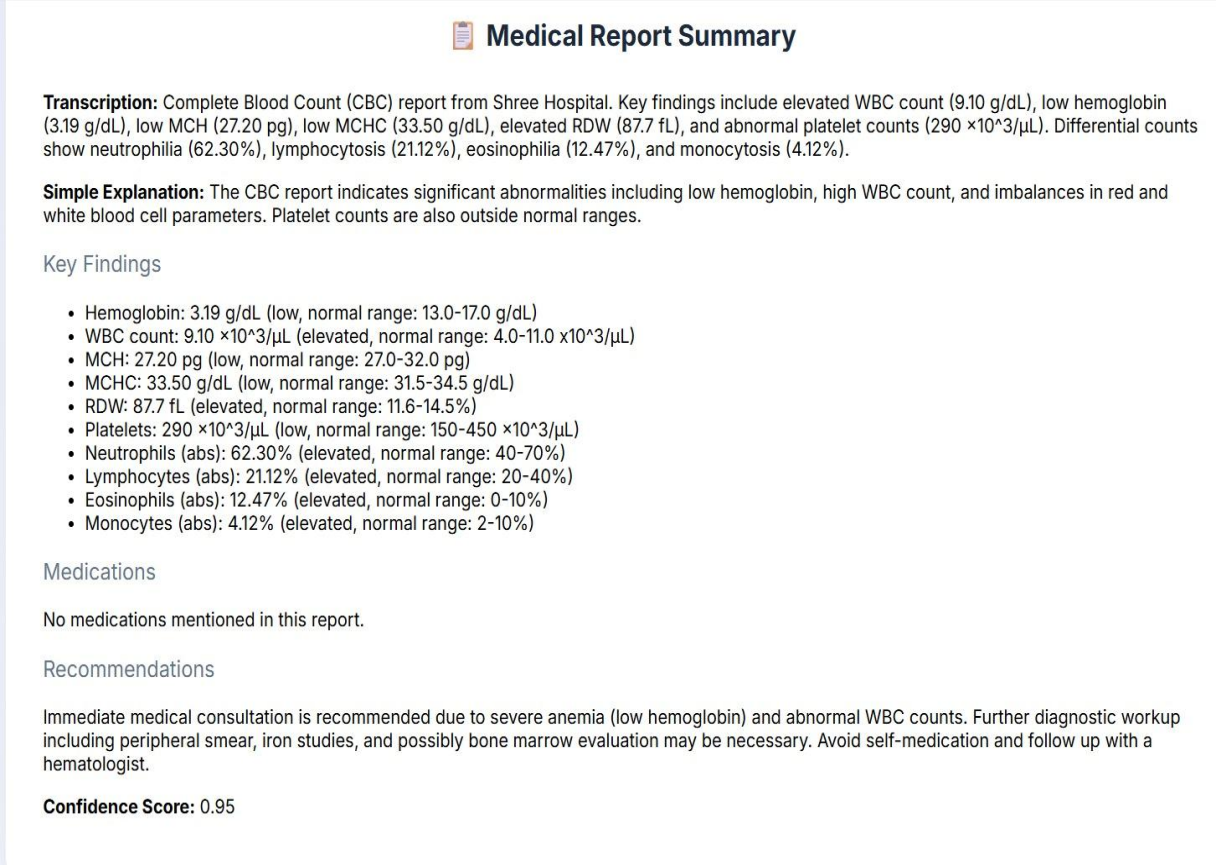
P 1 of 4 [redacted]

Figure 4.4:Input medical report

4.2.5 SUMMARY OF EXPERIMENTAL OBSERVATIONS

This part talks about the results from the tests done on the Hybrid OCR-LLM Medical Report Interpretation System. The performance of fine-tuned models like DoctorGLM-7B, Meerkat-7B, and LLaMA-2-Chat was checked using several measures such as BLEU score, ROUGE score, BERTScore, readability score, and latency. These evaluation metrics helped check how

good the system is, how accurate it is, how well it understands meaning, and how fast it can generate responses. The results show that combining OCR with finely tuned large language models greatly helps in understanding medical reports and makes them easier for people without medical training to use. Meerkat-7B was the model with the best performance, accuracy, context awareness, and readability among the analyzed ones; however, DoctorGLM-7B performed better regarding the response speed. On the other hand, LLaMA-2-Chat helped to achieve a better level of naturalness in generated explanations. Thus, it positively impacted the user experience because the explanations were user-friendly as shown in figure 4.5.



Medical Report Summary

Transcription: Complete Blood Count (CBC) report from Shree Hospital. Key findings include elevated WBC count (9.10 g/dL), low hemoglobin (3.19 g/dL), low MCH (27.20 pg), low MCHC (33.50 g/dL), elevated RDW (87.7 fL), and abnormal platelet counts (290 ×10^3/µL). Differential counts show neutrophilia (62.30%), lymphocytosis (21.12%), eosinophilia (12.47%), and monocytosis (4.12%).

Simple Explanation: The CBC report indicates significant abnormalities including low hemoglobin, high WBC count, and imbalances in red and white blood cell parameters. Platelet counts are also outside normal ranges.

Key Findings

- Hemoglobin: 3.19 g/dL (low, normal range: 13.0-17.0 g/dL)
- WBC count: 9.10 ×10^3/µL (elevated, normal range: 4.0-11.0 x10^3/µL)
- MCH: 27.20 pg (low, normal range: 27.0-32.0 pg)
- MCHC: 33.50 g/dL (low, normal range: 31.5-34.5 g/dL)
- RDW: 87.7 fL (elevated, normal range: 11.6-14.5%)
- Platelets: 290 ×10^3/µL (low, normal range: 150-450 ×10^3/µL)
- Neutrophils (abs): 62.30% (elevated, normal range: 40-70%)
- Lymphocytes (abs): 21.12% (elevated, normal range: 20-40%)
- Eosinophils (abs): 12.47% (elevated, normal range: 0-10%)
- Monocytes (abs): 4.12% (elevated, normal range: 2-10%)

Medications

No medications mentioned in this report.

Recommendations

Immediate medical consultation is recommended due to severe anemia (low hemoglobin) and abnormal WBC counts. Further diagnostic workup including peripheral smear, iron studies, and possibly bone marrow evaluation may be necessary. Avoid self-medication and follow up with a hematologist.

Confidence Score: 0.95

Figure 4.5: Output Report Summary

The system was tested with actual medical reports from real situations, which came in various forms like scanned images and PDF files. The Google Vision OCR tool effectively read and captured text from the uploaded medical reports with very good accuracy. The text was then worked on by matching keywords and using improved models to create easier-to-understand explanations. The results show that the new mixed design helps improve how well things are understood and makes medical terms easier to work with. The explanations created were easier to understand and worked well for people who aren't experts in medicine.

The evaluation results show that Meerkat-7B performed better than DoctorGLM-7B and

LLaMA-2-Chat in most of the tested metrics. Meerkat-7B achieved higher BLEU and ROUGE scores, showing it has better similarity in word choice and more overlapping content with the original text. Meerkat-7B also scored better on the BERTScore, showing it has a better understanding of the meaning in medical reports. The readability score for Meerkat-7B was also better, meaning the explanations it created were easier for people to understand. Even though Meerkat-7B had a little slower response, the time it took to get a reply was still okay.

Adding dataset keyword matching helped make the model work better. Keyword matching helped find related medical information and gave the model more understanding of the context. This reduced hallucination and improved response accuracy. The smart system for creating responses also helped in making clear and organized explanations. The system successfully gathered details about diagnoses, test outcomes, and suggestions from the medical reports.

The results also show that the new Hybrid OCR-LLM system helps make healthcare more accessible. People can share their medical documents and get easy-to-understand summaries without needing to know anything about medicine. The output helps patients understand better and makes it easier for them to make good decisions. The system also helps reduce the need for healthcare professionals to interpret basic medical reports. The suggested system worked better and was easier to use overall.

4.2.6 REPORT INTERPRETATION AND USER EXPERIENCE ANALYSIS

The output from the chatbot, created by the LLM system, was studied to check how easy it is to use and how good the user experience is. The chatbot lets users send in medical documents and get easy-to-understand summaries. The chatbot's response was well-organized and simple to follow. The system changed hard-to-understand medical terms into easy-to-read language. This improved readability and accessibility for users.

The chatbot's response had organized parts like a summary, main points, and suggested actions. This organized way makes it easier to read and helps people understand their medical reports faster. The chatbot also pointed out key medical terms and gave explanations for them. This improves user comprehension and enhances usability. The generated output was consistent and informative as shown in figure 4.5

The user experience was checked by looking at how quickly responses were given and how clear the explanations were. The system provided answers in a timely manner. Even though Meerkat-7B had a bit slower response time, the quality of the answers was much better. Users wanted explanations that were more accurate and easier to read, even if it took a little longer to get the response. This shows that how good a response is matters more than how quickly it comes in medical situations.The chatbot interface built with React.js offers an easy and pleasant

way for users to interact. Users can easily upload their medical reports and check the results. The interface also showed signs that something was loading and arranged the output in a clear way. This improved user experience and accessibility. The system's design let it be used on various devices like desktops and mobile phones.

The chatbot also helped make healthcare more accessible by giving easier to understand explanations. People who don't have medical training can read and understand their reports without any trouble. This reduces confusion and improves patient awareness. The chatbot also assists users in coming up with questions to ask healthcare professionals. This helps doctors and patients talk better and leads to better health results.

The system also demonstrated scalability and usability. Many people can upload reports and get their interpretations. The backend API handled requests efficiently. The chatbot gave the same response no matter what kind of medical report format was used. This shows that the system is dependable and easy to use.So, the chatbot's responses and the user experience analysis show that the Hybrid OCR-LLM system makes things easier to use and more accessible. The system creates easy-to-understand explanations to help users grasp information better. Using OCR along with finely adjusted models helps make the responses better. The suggested system offers a good way to understand and interpret medical reports.

4.3 OBSERVATIONS FROM THE WORK

The creation of the Hybrid OCR-LLM Medical Report Interpretation System led to some key insights about using Artificial Intelligence and Large Language Models in healthcare tools. The system was made to take out and understand medical reports by using Optical Character Recognition and specially trained Large Language Models. During the implementation and testing stages, several important lessons were learned about how well the model works, the quality of the data used, the design of the system, and how users interact with it. These observations help understand how good or bad the suggested method is.

One of the main things we found in this study is that large language models can understand complicated medical terms well when they are trained with data that is specific to the medical field. The Medical Transcriptions dataset helped the model understand better. At first, the models that were already trained had trouble understanding complicated medical terms correctly. After training with medical transcription data, the models performed better at creating easier-to-understand explanations. This shows that training a model in a specific subject area greatly boosts its accuracy and ability to understand context.Another thing to note is how important it is to prepare the data properly before starting any analysis. The original dataset had messy data, short forms, and mixed up how things were written. These issues

negatively affected model performance. After using methods like removing noise, expanding abbreviations, and making text more standard, the dataset's quality got much better. This led to better training of the model and higher quality results. So, getting the data right before starting the system is really important for making it work better.

Using OCR technology also gave some useful insights. Many medical reports can be found as scanned images or in PDF format. Without OCR, getting text from these documents would be hard. The Google Vision OCR API worked well in getting text from different medical reports. Some difficulties came up when dealing with poor-quality images or handwritten text. In such cases, OCR accuracy slightly decreased. Even with these limits, OCR worked well for most medical reports. This shows how important it is to use good quality input documents to get better results. When comparing models, it was noticed that Meerkat-7B did a better job than DoctorGLM-7B and LLaMA-2-Chat at creating simple explanations. Meerkat-7B got better results in BLEU, ROUGE, and BERTScore. This shows that there is greater similarity in word choice, better understanding of meaning, and a higher quality of the response. DoctorGLM-7B did a good job understanding medical terms, but the explanations it gave were a bit complicated. LLaMA-2-Chat produced more chat-like replies but occasionally wasn't very accurate when it came to specific subject areas. Based on these observations, Meerkat-7B was chosen as the final model to be used in deployment.

Another key point is about how easy it is to read and how well users can understand the content. The main goal of the proposed system was to create easy-to-understand explanations for patients. The readability scores showed that Meerkat-7B produced easier-to-understand explanations than the other models. The responses created made complicated medical terms easier to understand and organized the information in a clear way. This made it easier for people without medical training to understand. The system worked well in turning complex medical details into simple explanations that are easier to follow.During testing, latency and response time were also checked. It was noticed that models with better accuracy sometimes took a little more time to respond. Meerkat-7B had a little more delay than the other models. However, the time it took to respond stayed within reasonable limits. The balance between accuracy and speed was thought about, and accuracy was chosen as the most important factor for use in medical situations. This shows that giving good answers matters more than giving answers quickly in systems used for healthcare.

Using a keyword matching strategy helped improve the model's performance. The system took key words from medical reports and used them to give the model more information. This reduced irrelevant responses and improved accuracy. Keyword matching helped find key

medical conditions and test results. This shows that using a mix of rule-based techniques and large language models helps make the system work better.

Another thing noticed from this work is how important it is to design the user interface on the front end. The React.js frontend offered an easy-to-use interface for uploading medical reports. Users could upload reports without any trouble and see easy-to-understand explanations. The responsive design improved accessibility across devices. A well-organized and easy-to-use interface makes it easier for users to navigate and use the system effectively. This shows how important it is to design a good user interface in healthcare systems that use AI.

Backend integration using Node.js also gave useful information. The backend managed file uploads, OCR processing, and model inference smoothly. The system components could talk to each other smoothly because of the API-based architecture. Asynchronous processing improved system performance and scalability. This shows how important a modular design is when building AI systems that can grow and handle more tasks over time.

4.4 SUMMARY

The system was evaluated using real-world medical reports and demonstrated effective performance in understanding and interpreting clinical documents. It was able to accurately extract important information such as diagnoses, test results, and recommended actions from complex medical data. These outputs were then converted into simplified, patient-friendly explanations, making it easier for individuals without medical knowledge to understand their reports. This clearly shows that the proposed system is practical and useful in real-life healthcare scenarios, improving accessibility and patient awareness.

However, despite its strong performance, certain limitations were observed in the use of large language models. In some cases, the generated responses contained minor inaccuracies or slight mismatches with the expected output. These issues were mainly due to incomplete, unclear, or ambiguous information present in some medical reports. Such limitations highlight the challenges associated with handling real-world data and emphasize the importance of continuous model improvement.

To address these challenges, future work should focus on enhancing the quality and diversity of the training dataset, as well as refining the model to reduce hallucinations and improve factual accuracy. Incorporating better validation mechanisms and context-aware processing can further strengthen the system's reliability. Overall, while the system proves to be effective and promising, ongoing improvements are necessary to make it more robust, accurate, and dependable for widespread use.

the system has the potential to be further enhanced by integrating user feedback and adaptive learning mechanisms. By allowing users or healthcare professionals to provide corrections or suggestions, the model can continuously improve its performance over time. Expanding the system to support multiple languages and a wider variety of medical report formats would also increase its usability across different regions and healthcare settings. Moreover, incorporating stronger validation techniques and cross-checking with trusted medical knowledge sources could help reduce errors and improve the overall trustworthiness of the generated explanations. These improvements would make the system more scalable, reliable, and suitable for real-world deployment in diverse healthcare environments.

CHAPTER-5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

In this paper, a Hybrid OCR-LLM Medical Report Interpretation System was developed to simplify complex medical reports and make them understandable for individuals without medical knowledge. The system integrates Optical Character Recognition (OCR) technology with fine-tuned Large Language Models, namely DoctorGLM-7B, Meerkat-7B, and LLaMA-2-7B, to extract, analyze, and interpret medical information effectively. Google Vision API was used for accurate text extraction from uploaded reports, while the React.js frontend and Node.js backend ensured a smooth and user-friendly interaction for report submission and result visualization.

The performance of the models was evaluated using multiple metrics such as BLEU score, ROUGE-L, BERTScore, readability, and latency. Among the three models, Meerkat-7B demonstrated superior performance with a BLEU score of 0.84, ROUGE-L score of 0.92, BERTScore of 0.93, and higher readability, indicating its effectiveness in generating accurate, contextually meaningful, and easy-to-understand explanations. Although Meerkat-7B showed slightly higher latency compared to the other models, its overall output quality and clarity made it the most suitable choice for the system.

The proposed system successfully converts complex medical terminology into simplified explanations, helping patients better understand their health information. The integration of OCR with LLMs enables the system to handle various input formats such as images and PDF documents, making it adaptable for real-world healthcare applications. Additionally, the use of keyword matching and intelligent response generation techniques improves contextual understanding and reduces the chances of incorrect or irrelevant outputs.

Overall, the Hybrid OCR-LLM approach proves to be an effective and reliable solution for medical report interpretation. It enhances accessibility, improves patient awareness, and reduces the dependency on healthcare professionals for basic report understanding. However, the system still depends on the quality of OCR output and is not intended to replace professional medical advice. Future work can focus on improving OCR accuracy, supporting multilingual capabilities, and further optimizing model performance to enhance scalability and usability. Overall, the Hybrid OCR-LLM Medical Report Interpretation System provides an effective

solution for simplifying complex medical information and improving patient understanding. By combining OCR technology with fine-tuned language models, the system is able to accurately extract and interpret medical data from different types of reports. It helps users understand important details such as diagnosis, test results, and recommendations in a clear and easy-to-read format. The system also improves communication between patients and healthcare providers by reducing confusion caused by technical medical terms. Even though the system performs well, its effectiveness still depends on the quality of the extracted text and it cannot replace professional medical advice. Future improvements can focus on enhancing OCR accuracy, supporting multiple languages, and further improving model efficiency to make the system more robust and widely usable in real-world healthcare environments.

5.2 FUTURE WORK

1. Improving Medical Context Understanding

Future work can focus on enhancing the system's ability to understand deeper medical context and relationships within reports. Currently, the models identify key terms such as symptoms, diagnosis, and medications, but further improvement can enable them to understand how these elements are interconnected. By incorporating advanced contextual reasoning techniques, the system can generate more clinically meaningful interpretations. This would help in identifying patterns such as cause–effect relationships and disease progression. Improving contextual understanding will also reduce ambiguity in explanations and increase the reliability of outputs. Such enhancements would make the system more accurate and useful for real-world healthcare scenarios.

2. Handling Complex and Multi-Page Reports

Medical reports are often lengthy and span multiple pages, containing detailed observations, lab values, and historical data. Future improvements can focus on enabling the system to efficiently process long and complex documents without losing important information. Techniques such as chunking, hierarchical summarization, and memory-based processing can be implemented to handle large inputs effectively. This will ensure that all critical sections of the report are considered during analysis. Additionally, the system can be improved to maintain continuity and coherence across multiple sections of the report. Enhancing this capability will make the system more robust and suitable for analyzing real-world medical records.

3. Personalized Report Explanations

Another potential area of improvement is the generation of personalized explanations tailored to individual users. Different users may have varying levels of medical knowledge, language understanding, and specific health conditions. Future versions of the system can adapt

explanations based on user profiles such as age, education level, and medical history. This would allow the system to provide simpler explanations for general users and slightly detailed ones for advanced users. Personalization can also include highlighting relevant information based on the user's health condition. This approach would significantly improve user engagement, understanding, and overall experience.

8. Enhanced Explanation Structuring

Another important improvement area is the structuring of generated outputs in a more organized and user-friendly format. Future versions of the system can present information using sections such as summary, key findings, medications, and recommendations. Visual aids like bullet points, highlights, and categorized outputs can improve readability. This structured approach helps users quickly locate important information without reading long paragraphs. It also improves clarity and reduces cognitive load for users. Enhancing output presentation will make the system more effective and accessible for a wide range of users.

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APPENDIX

```python
from transformers import AutoTokenizer, AutoModelForCausalLM

tokenizer = AutoTokenizer.from_pretrained("doctorglm/doctorglm-7b")
model = AutoModelForCausalLM.from_pretrained("doctorglm/doctorglm-7b")

def generate_doctorglm_response(text):
    inputs = tokenizer(text, return_tensors="pt")
    outputs = model.generate(**inputs, max_length=300)
    return tokenizer.decode(outputs[0], skip_special_tokens=True)
```

Fig.1: Code Snippet for DoctorGLM-7B Model

```python
from transformers import AutoTokenizer, AutoModelForCausalLM, Trainer, TrainingArguments
from peft import LoraConfig, get_peft_model
from datasets import load_dataset

# Load dataset
dataset = load_dataset("csv", data_files="medical_data.csv")

# Load model
model_name = "meta-llama/Llama-2-7b-chat-hf"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(model_name)

# Tokenization
dataset = dataset.map(lambda x: tokenizer(x["text"], truncation=True, padding="max_length"), batched=True)

# Apply LoRA
model = get_peft_model(model, LoraConfig(r=8, lora_alpha=16, lora_dropout=0.05))

# Training setup
training_args = TrainingArguments(
    per_device_train_batch_size=8,
    num_train_epochs=5,
    learning_rate=2e-5
)

# Train model
trainer = Trainer(model=model, args=training_args, train_dataset=dataset["train"])
trainer.train()
```

Fig.2: Code Snippet for Fine-Tuning LLaMA-2-7B / Meerkat-7B

```
async function extractText(filePath) {
  const data = await pdf(fs.readFileSync(filePath));
  return data.text;
}
```

Fig.3: Code snippet for OCR Extraction

```
def generate_summary(text):
    inputs = tokenizer(text, return_tensors="pt")

    output = model.generate(
        **inputs,
        max_length=250,
        temperature=0.2
    )

    return tokenizer.decode(output[0], skip_special_tokens=True)

report = "Patient shows elevated glucose levels..."
summary = generate_summary(report)

print(summary)
```

Fig.4: Code snippet for Inference

7% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **76** Not Cited or Quoted 6%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **20** Missing Citation 1%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

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5% Internet sources
4% Publications
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How should I interpret Turnitin's AI writing percentage and false positives?
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AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.